

Lab 1: Adobe Photoshop Basics

♦ **Title:** Basic Overview of Adobe Photoshop

♦ **Objective:**

The main objective of this lab is to understand the basic tools and features of Adobe Photoshop, including the tools panel and layers panel, and to perform basic image editing and adjustments.

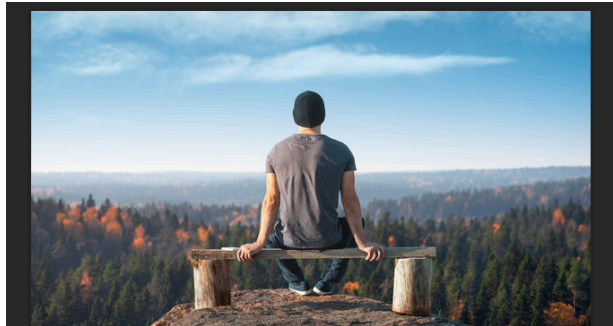
♦ **What We Did:**

- Explored **Tools Panel** and **Layers Panel**
- Performed **image adjustments**:
 - Brightness & Contrast
 - Vibrance
 - Hue/Saturation
 - Black & White conversion
- Worked with **text and shapes on images**
- Saved images in different **file formats (PSD, JPEG, PNG, etc.)**
- Compared **image quality and file sizes**

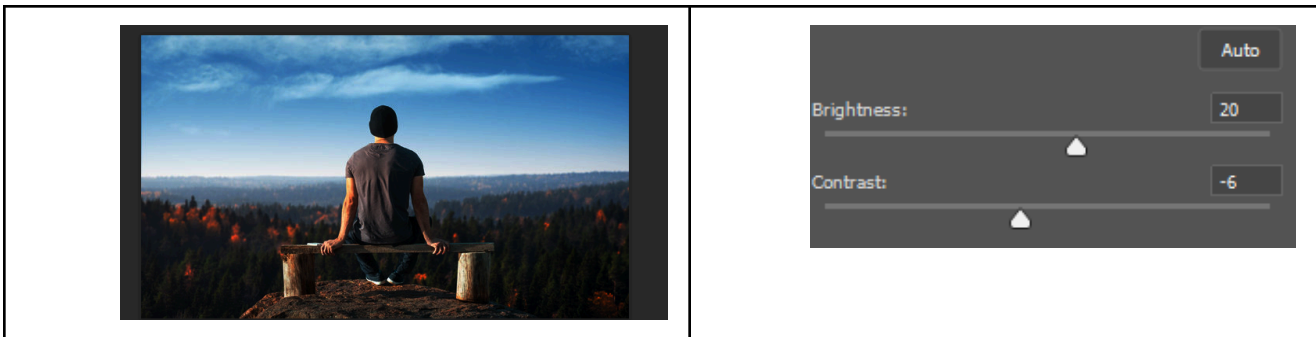
♦ **Tools/Software Used:**

- Adobe Photoshop

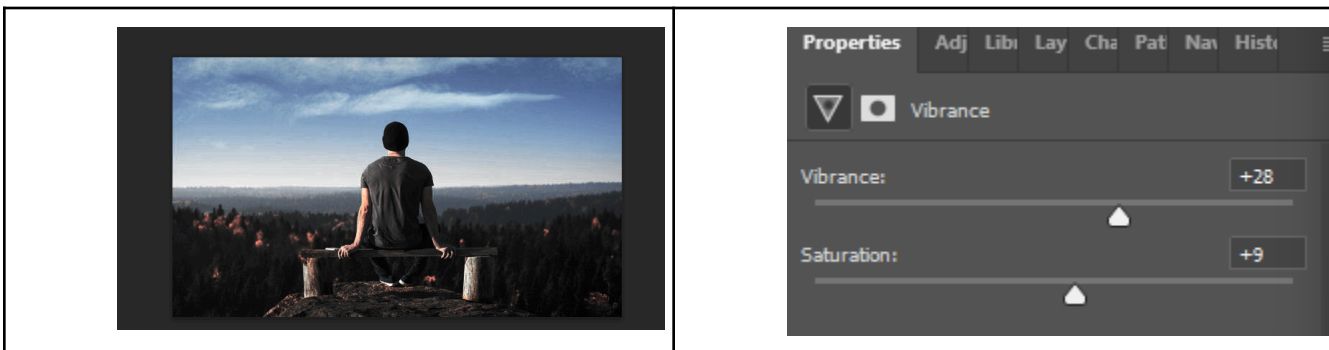
- Original image



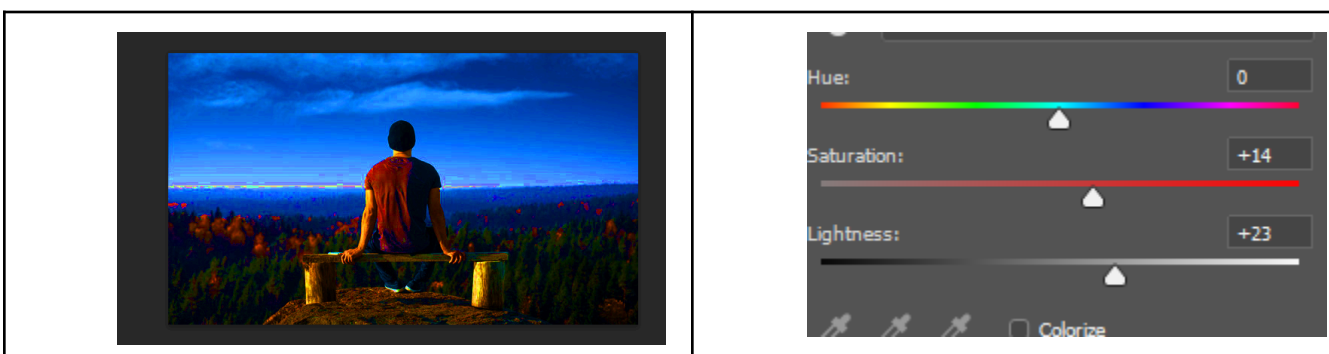
- Brightness and contrast



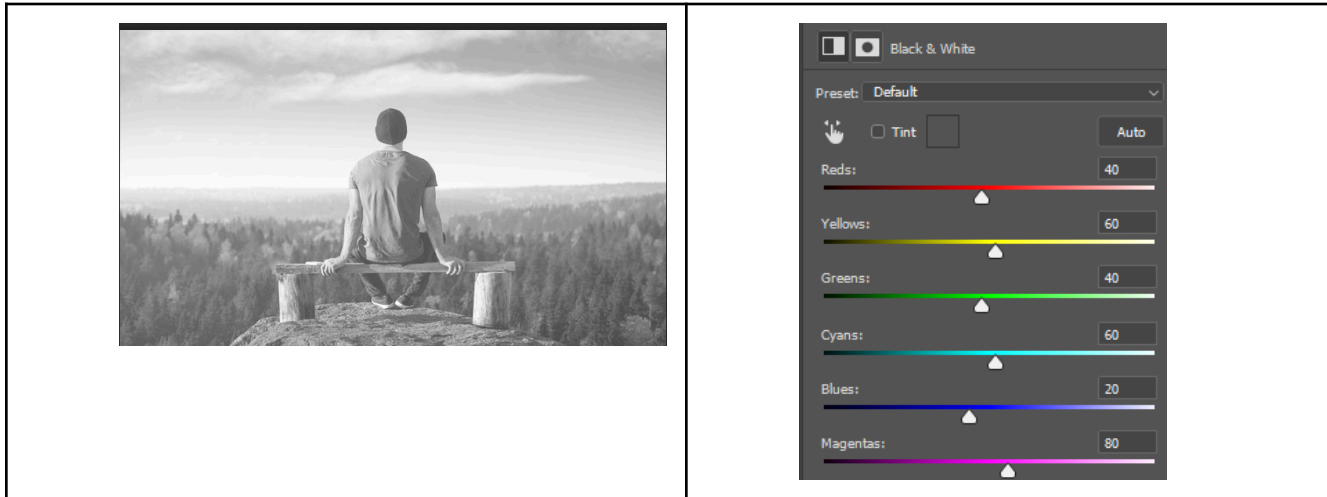
- Color Vibrance



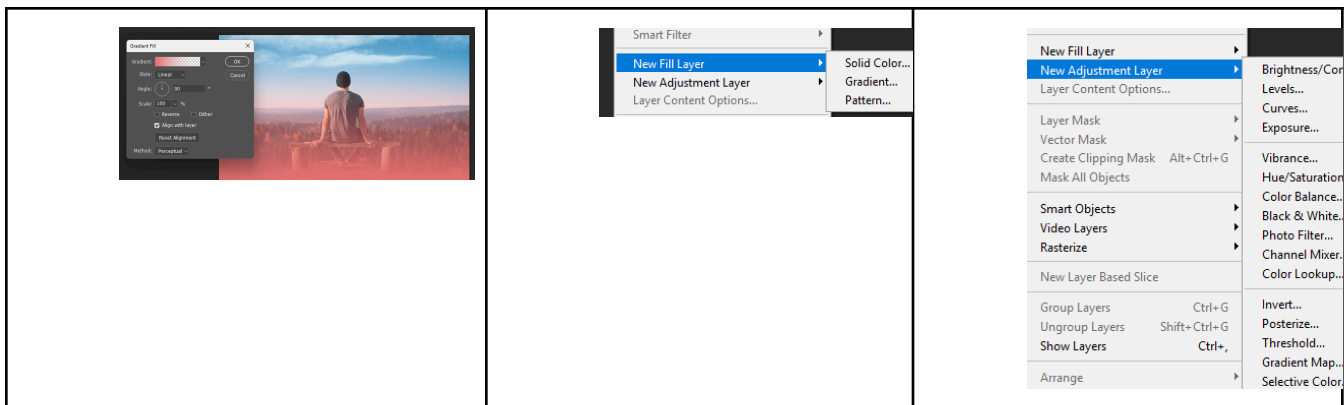
- Hue, Saturation and Lightness



- Convert color image into black and white image

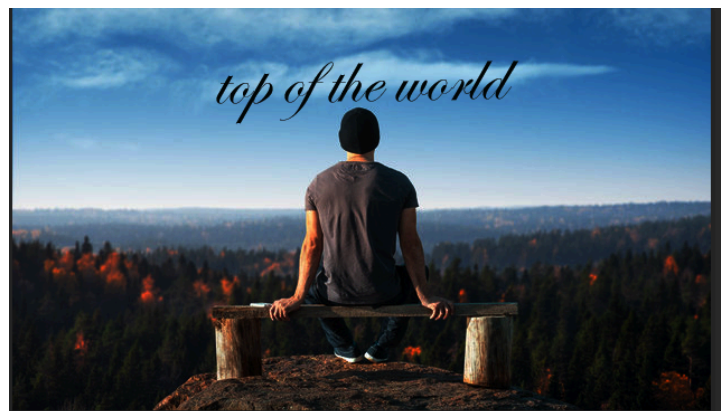


- Adjustment and fill layers



Experiment 3: Working with text and shapes on image

Select one image and use different text in that image (as your choice of design, font, font size and style). Make changes on that image such as contrast, light so that added text, shapes overshadow the image like some announcement in text form and image as a background.



Lab 2: Advanced Photoshop & Design

♦ **Title:** Image Retouching and Creative Design

♦ **Objective:**

To learn advanced editing techniques such as retouching, background manipulation, morphing, and creative design for event promotion.

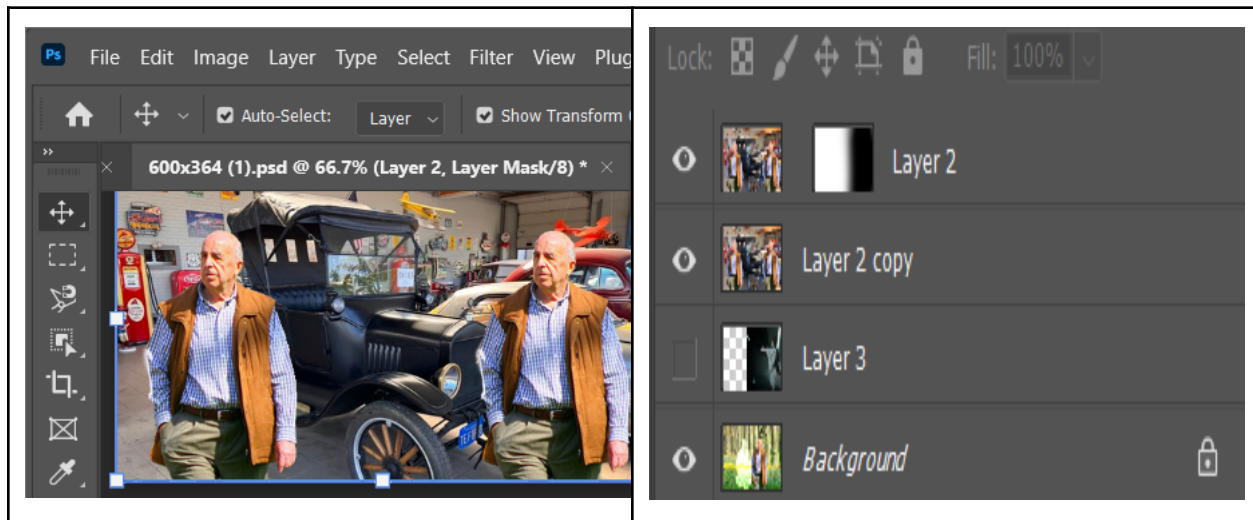
♦ **What We Did:**

- Retouched images:
 - Removed objects
 - Cloned areas
 - Applied blur using smart filters
- Combined multiple images using:
 - Layer masks
 - Background replacement
- Performed **image morphing**
- Designed:
 - Event Logo
 - Social Media Banner

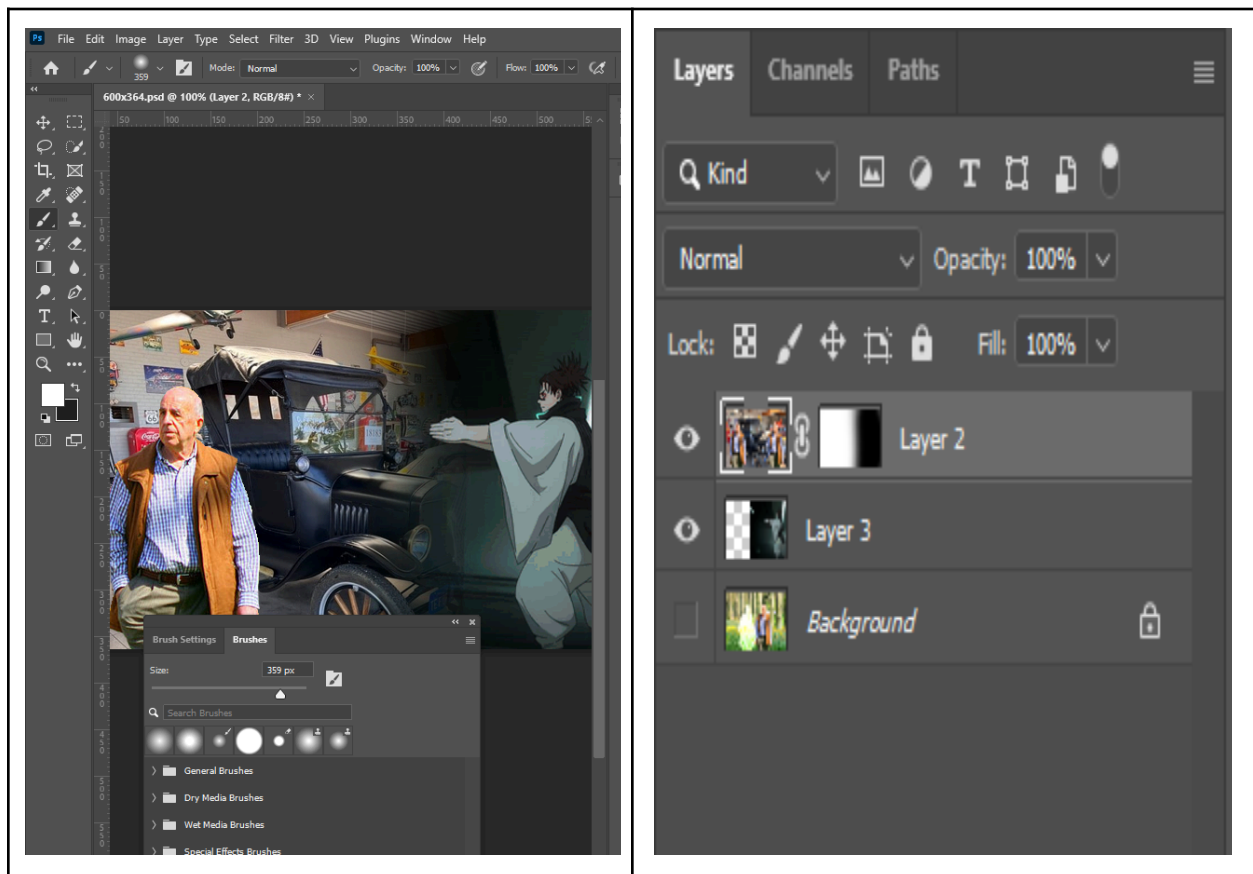
♦ **Tools/Software Used:**

- Adobe Photoshop

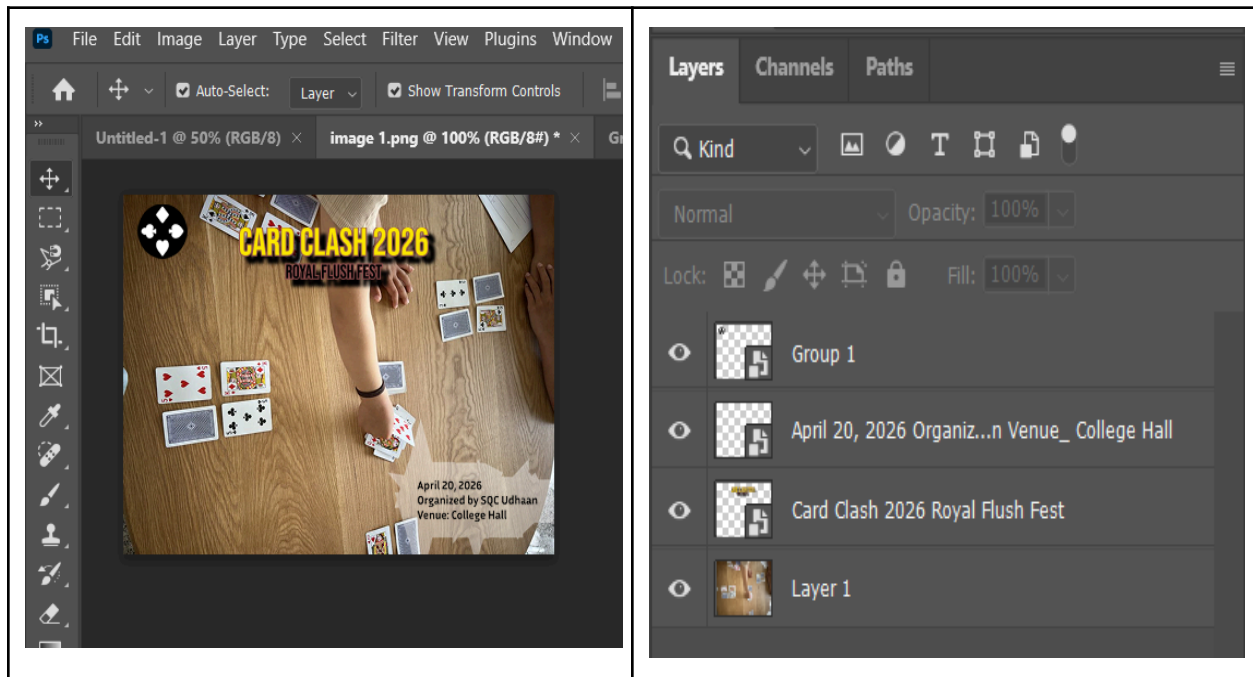
Layer used for Experiment 2: Working with more than one image



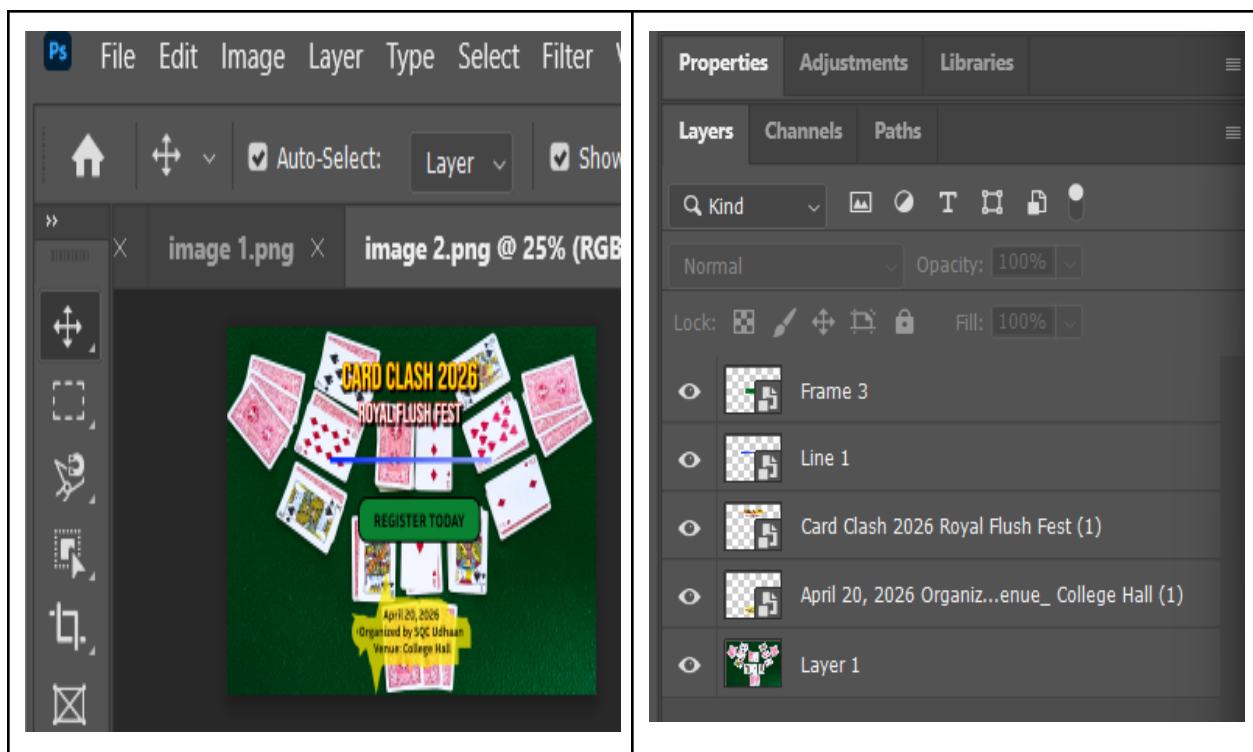
Layer used for Experiment 3: Morphing technique



Layer used for Experiment 4: Event logo with event name, date and relevant graphic



Layer used for Experiment 4: Social media graphics/Banner to promote the event



Lab 3: Figma UI/UX Design

♦ **Title:** Basic Overview of Figma for Design Applications

♦ **Objective:**

To understand the basics of UI/UX design using Figma, including frames, components, and prototyping.

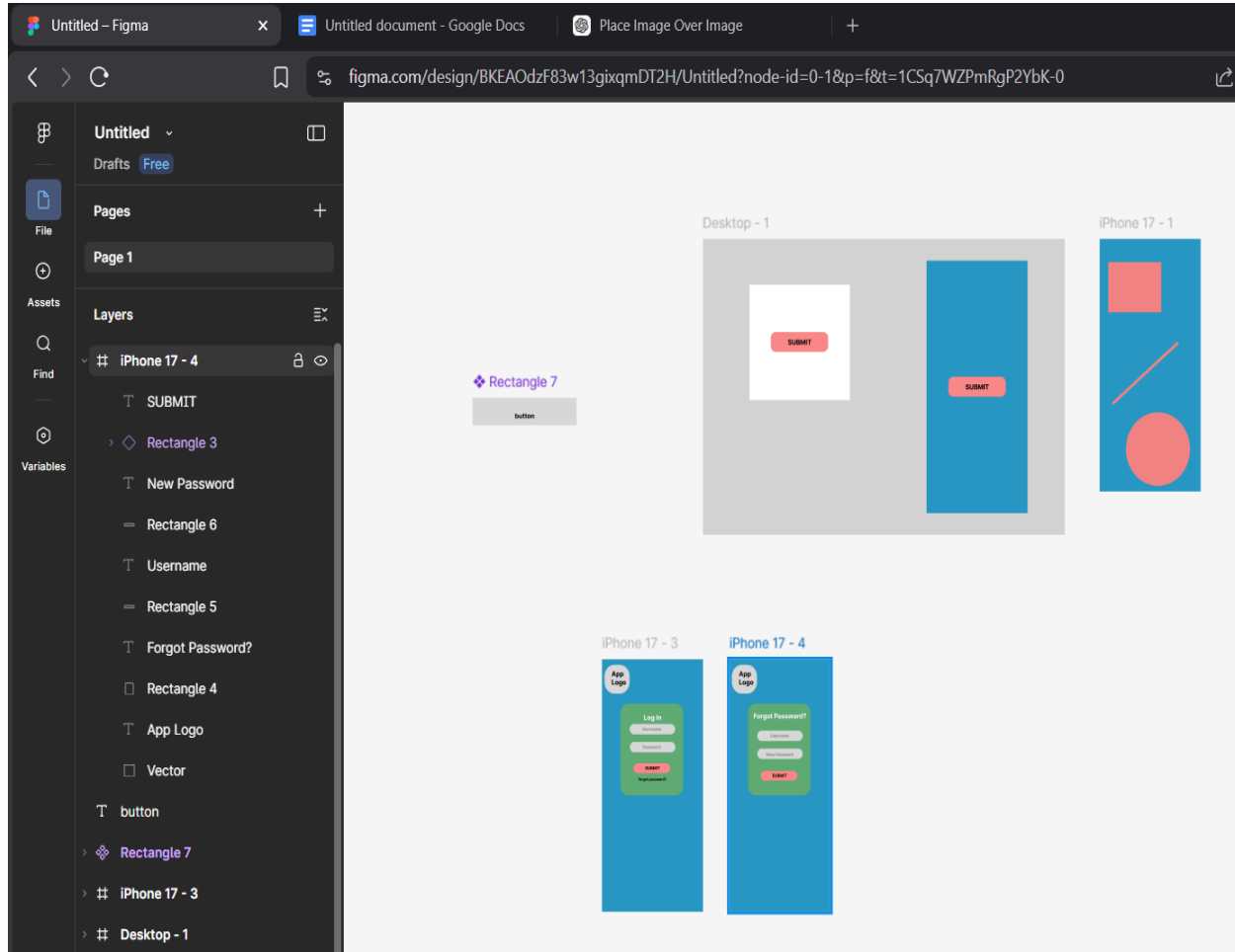
♦ **What We Did:**

- Explored:
 - Layers Panel
 - Properties Panel
 - Toolbar
- Created:
 - Mobile frame
 - Basic shapes
- Designed:
 - Reusable button component
 - Login screen UI
- Built prototype:
 - Linked screens (Login → Forgot Password)
 - Added transitions

♦ **Tools/Software Used:**

- Figma

Layer used in this lab 3



Lab 4: Advanced Image Editing and Design in Adobe Photoshop

♦ **Title:** Advanced Image Editing and Creative Design using Adobe Photoshop

♦ **Objective:**

The main objective of this lab is to learn advanced image editing techniques such as retouching, background replacement, morphing, and creative design for event promotion using Adobe Photoshop.

♦ **What We Did:**

Performed image retouching:

- Used selection tools to edit specific parts of an image
- Applied Smart Filters to blur images
- Removed unwanted objects from images
- Added objects using cloning techniques

Worked with multiple images:

- Combined two or more images
- Replaced backgrounds using layer masks
- Set one image as the background of another

Applied morphing techniques:

- Blended two images smoothly to create a single combined image

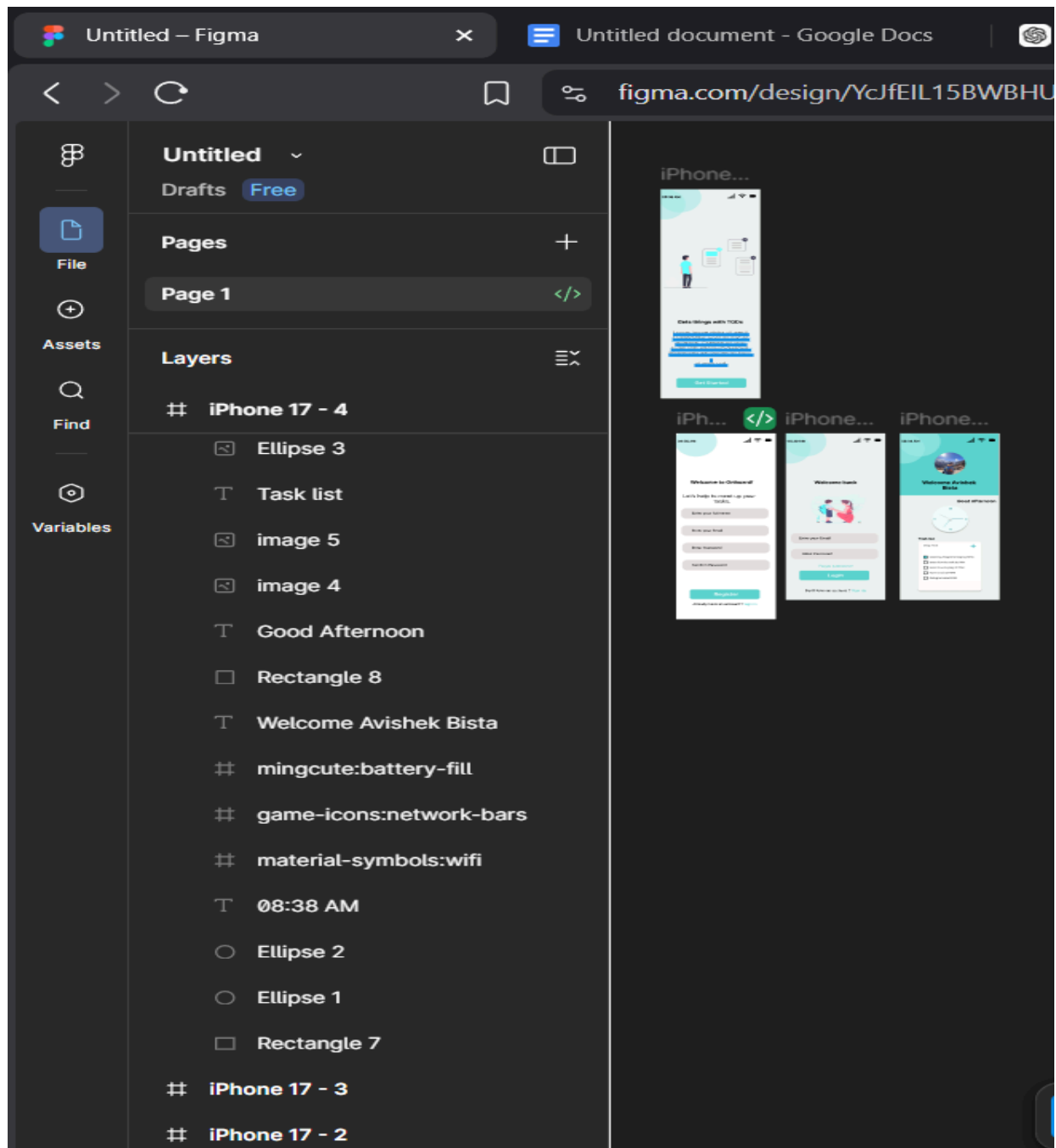
Designed creative content for an event:

- Created an event logo representing the theme
- Designed social media graphics/banner for promotion

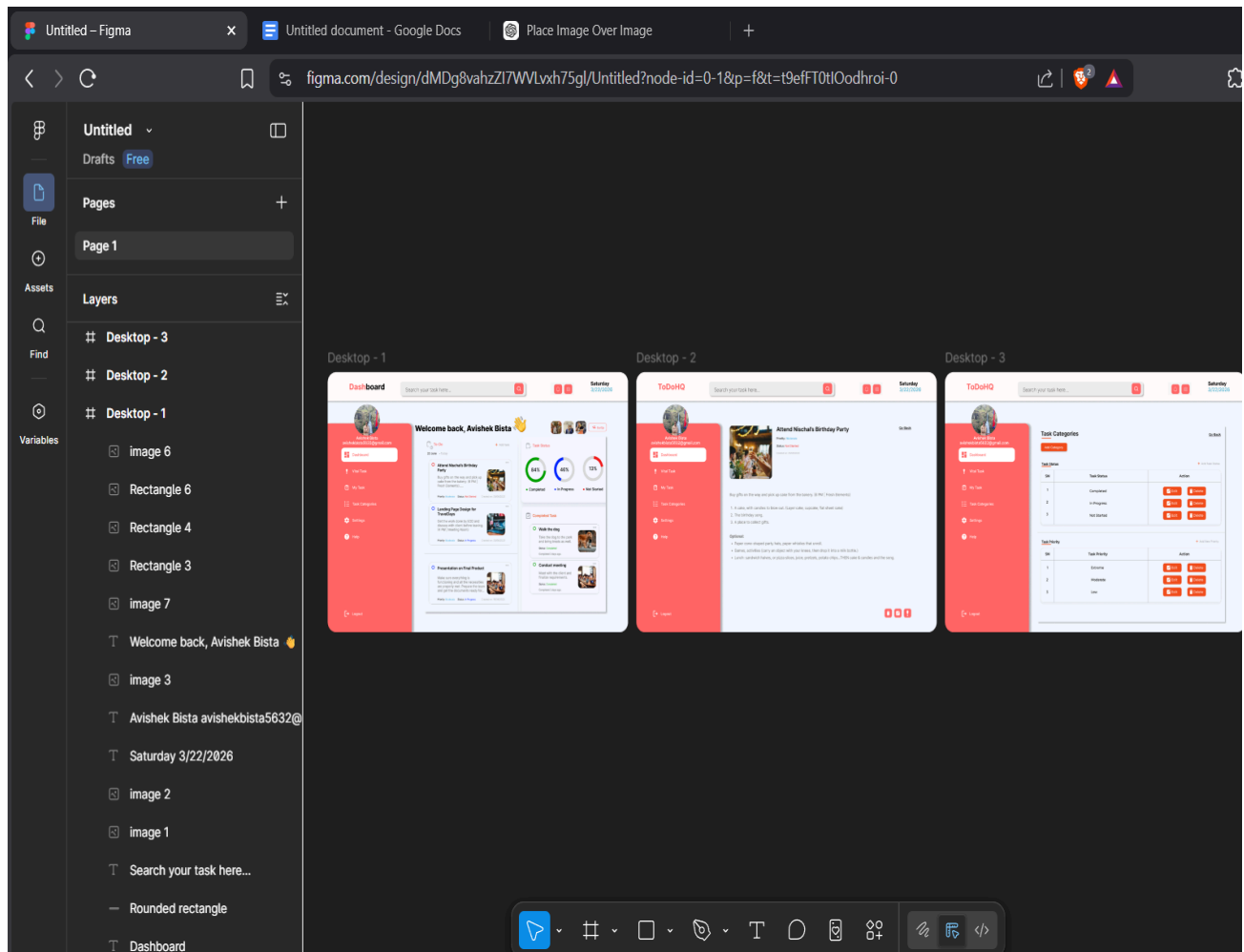
♦ **Tools/Software Used:**

Adobe Photoshop

Layer used in lab 4 for figma app design



Layer used in lab 4 for figma website design



Lab 5: Adobe Illustrator Basics and Vector Design

♦ Title:

Basic Overview and Design using Adobe Illustrator

♦ Objective:

The main objective of this lab is to understand the basic tools and features of Adobe Illustrator and to create vector-based designs using shapes, text, and tracing techniques.

♦ What We Did:

Explored Illustrator interface:

- Studied tools panel and layers panel
- Worked with color combinations and gradient features
- Created random artwork using different tools

Designed shapes:

- Created basic shapes
- Combined multiple shapes to form complex designs

Worked with text:

- Used text tools to create designs
- Applied effects like curved text, outline text, shadow text, wrap text, and 3D effects

Performed image tracing:

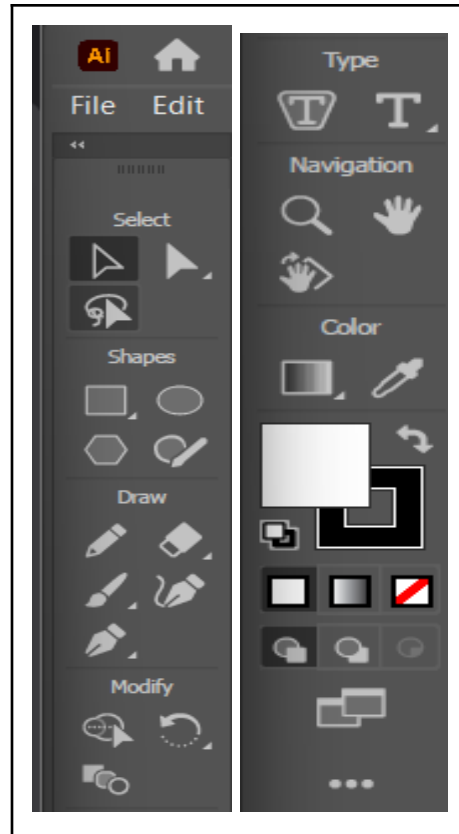
- Selected an image and traced its outline
- Converted raster image into vector drawing

♦ Tools/Software Used:

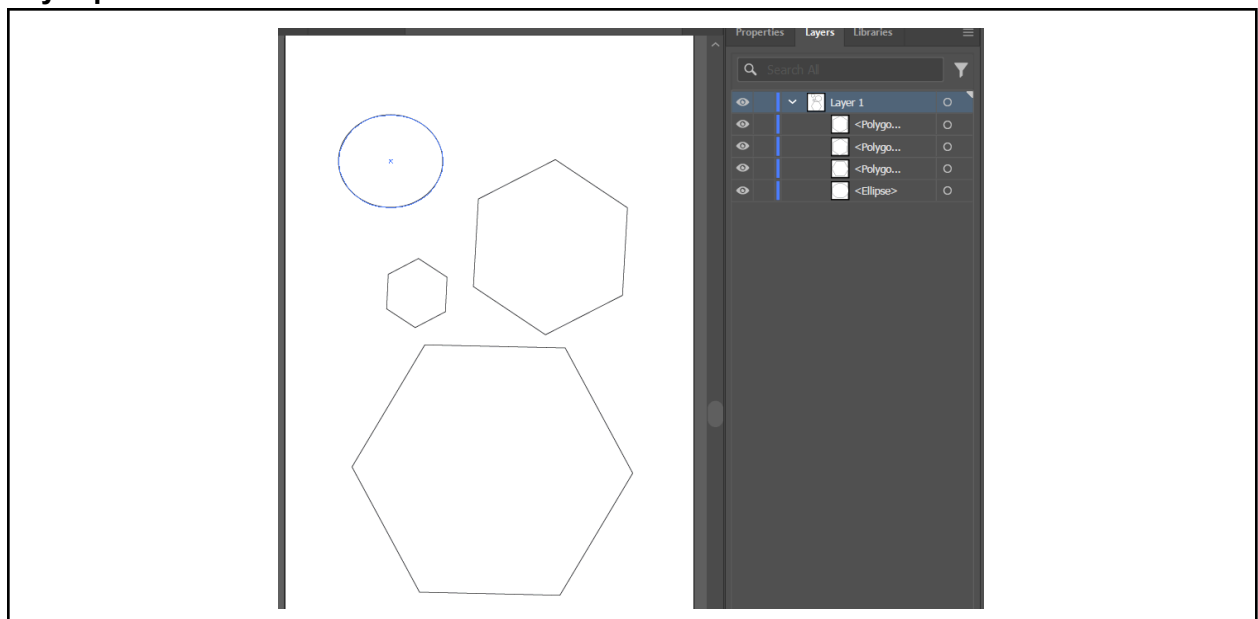
Adobe Illustrator

Experiment 1: Basic Overview of Adobe Illustrator Focus on tools panel, layers panel, color combination and gradients features. (You may perform random artwork with different tools panels with color combinations)

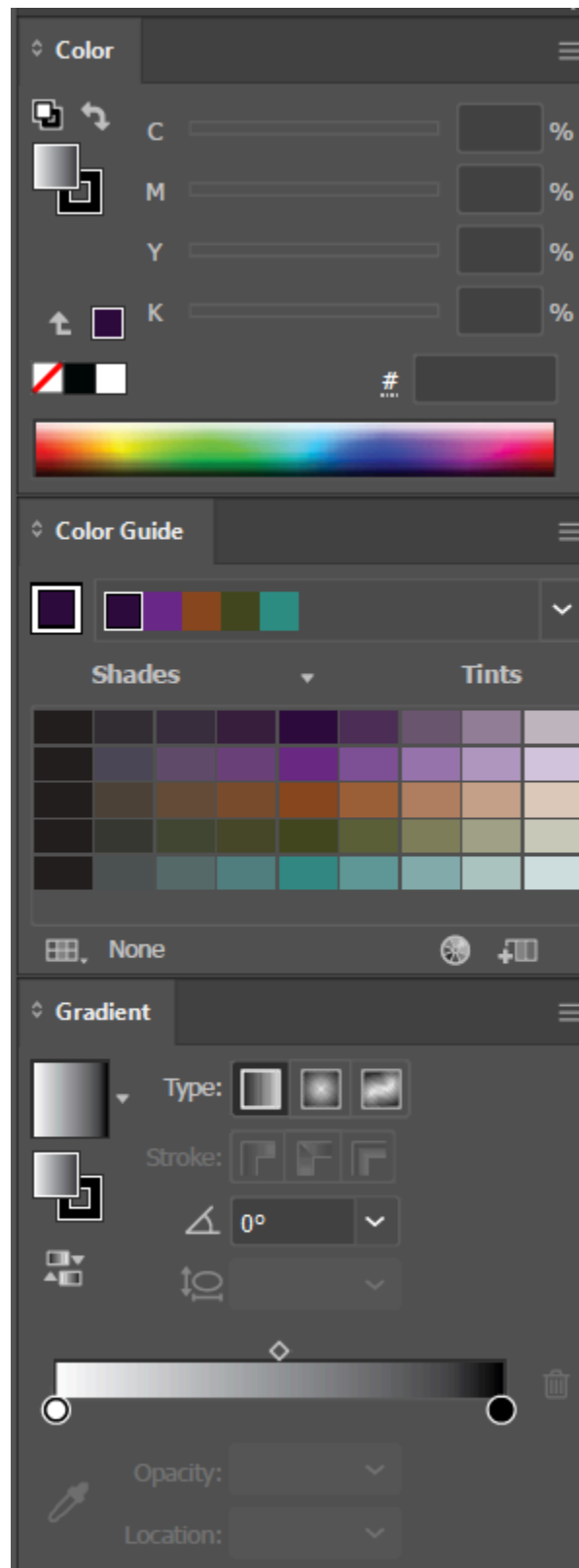
Tool panel:



Layer panel:

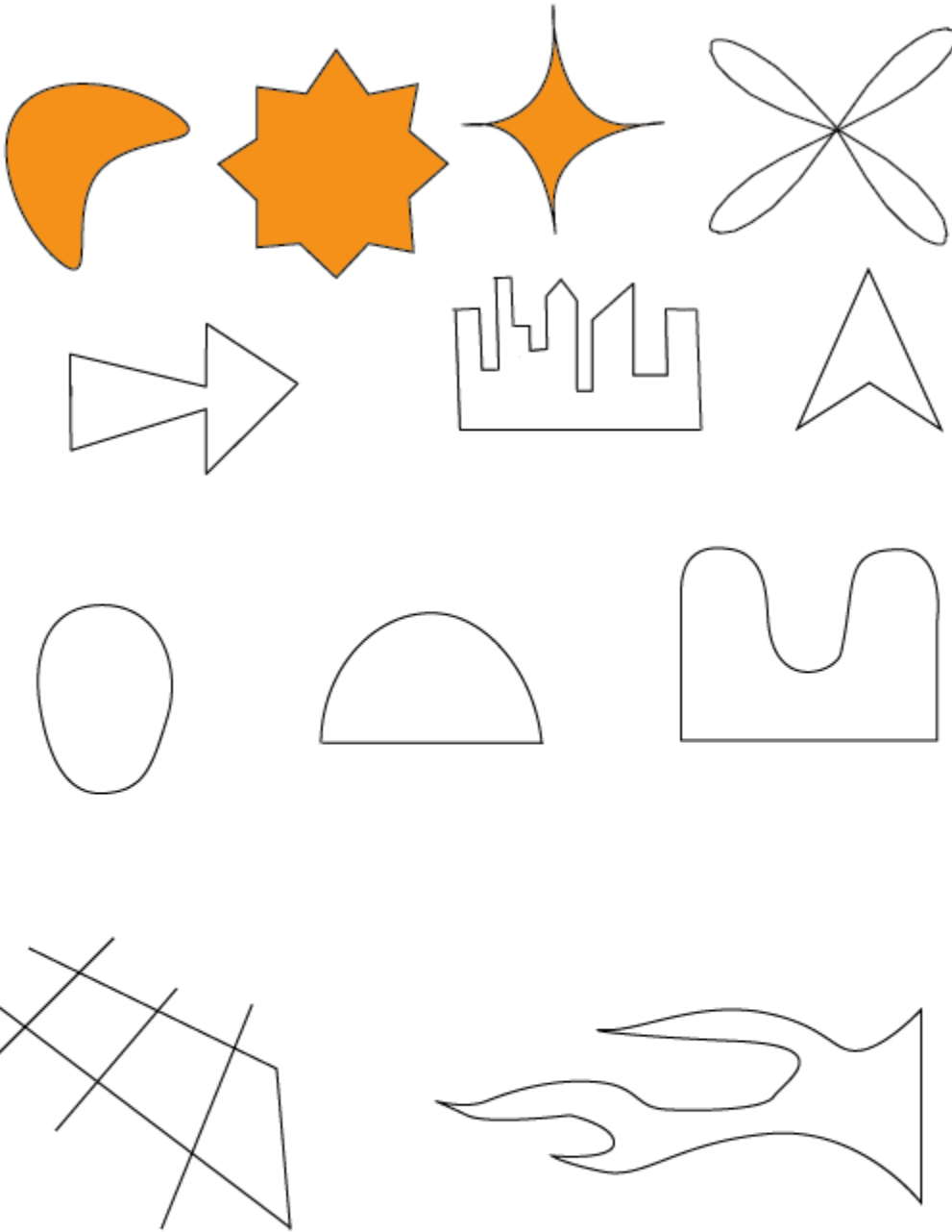


Color combination and gradients features:

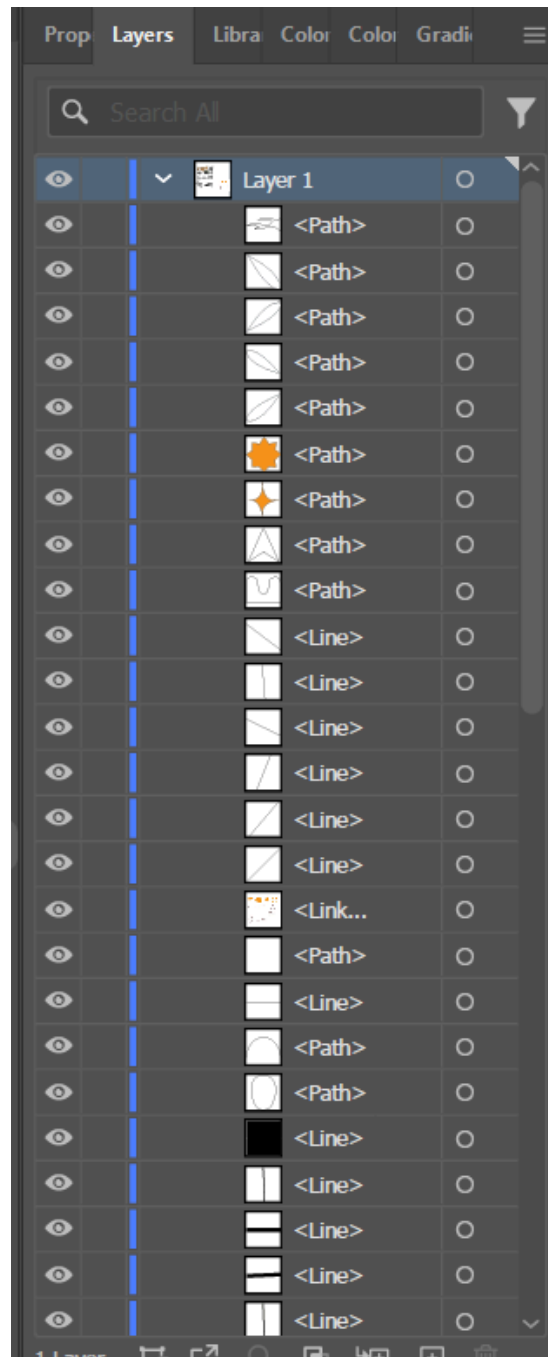


Experiment 2: Design basic shapes Design some basic shapes provided in lab class.
Combine multiple shapes as provided in lab class.

Basic shapes:



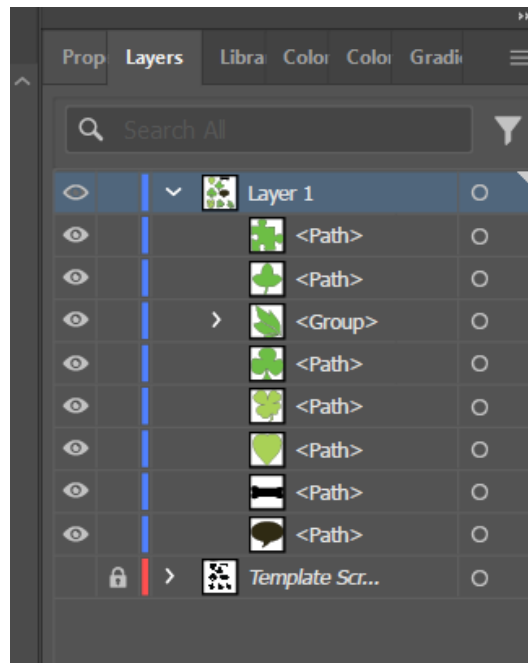
Layers:



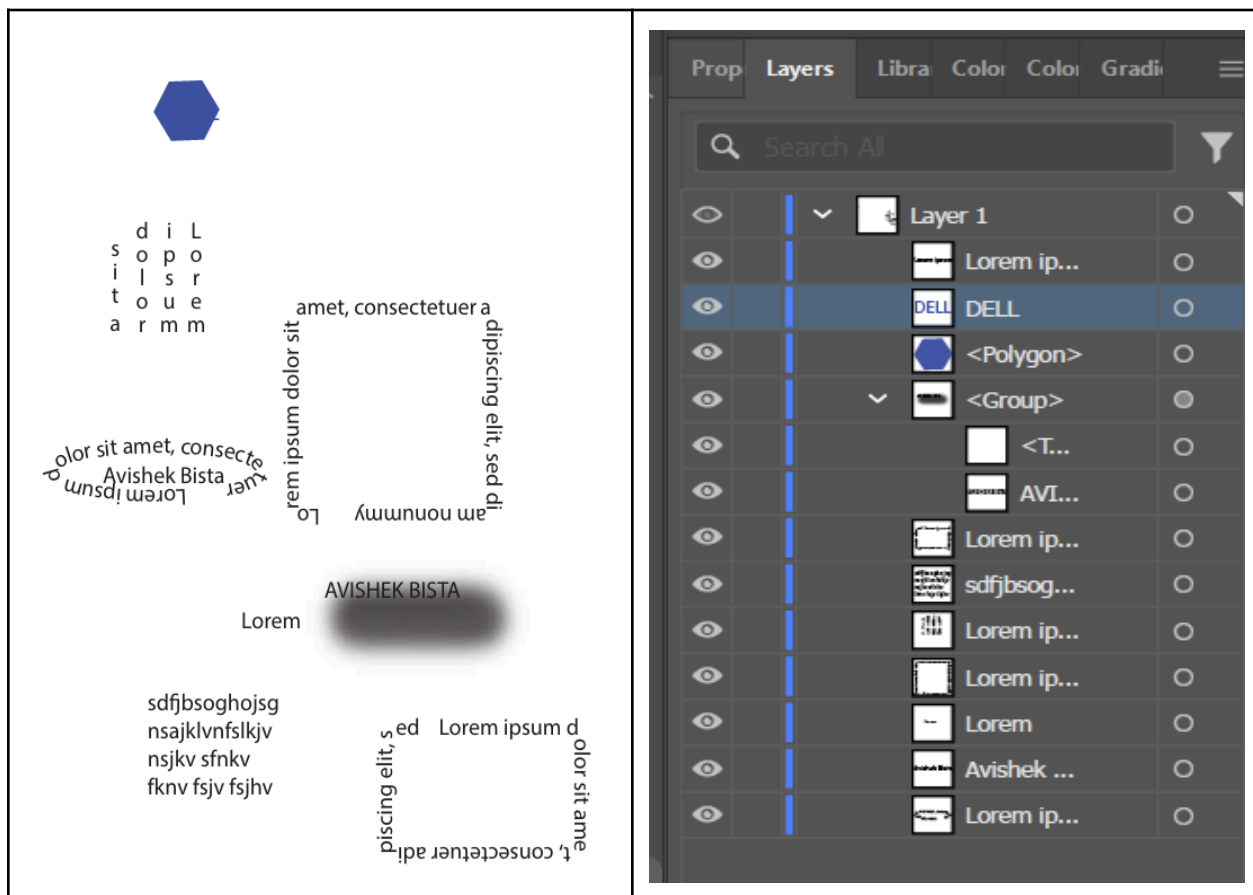
Combinations of shapes:



Layers:



Experiment 3: Working with text Use text tools and perform different designs on that text such as curved text, outline text, shadow text, wrap text and 3D effect.]



Experiment 4: Tracing digital image Choose a random image and trace out that image outline to convert into vector drawing.

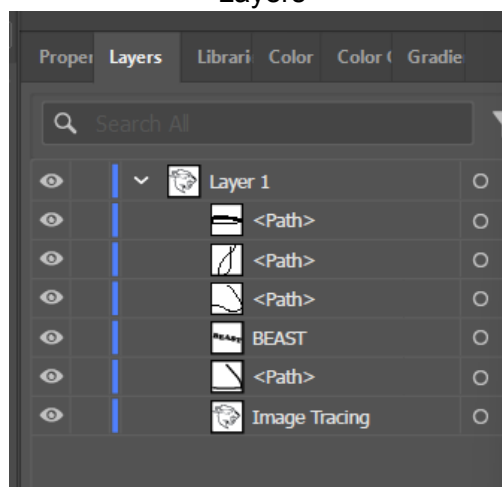
Original Image



Vector Image



Layers



Lab 6: Advanced Illustrator Design and Creative Projects

♦ **Title:**

Creative Design using Adobe Illustrator

♦ **Objective:**

The main objective of this lab is to enhance design skills by creating shapes, icons, and promotional materials using Adobe Illustrator.

♦ **What We Did:**

Worked with shapes:

- Designed shapes provided in the lab

Created icons:

- Designed 5 basic icons of different types

Designed promotional material:

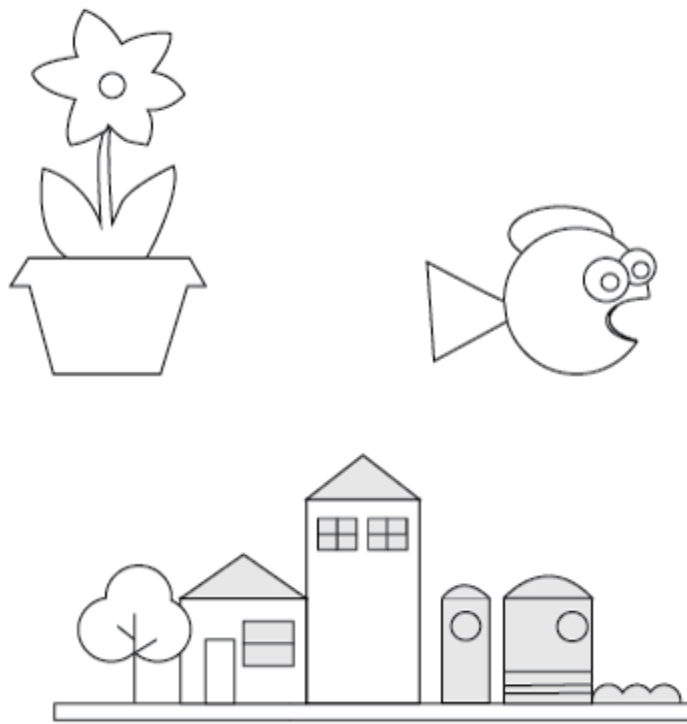
- Created a flyer/leaflet for an information session
- Focused on layout, text arrangement, and visual appeal

♦ **Tools/Software Used:**

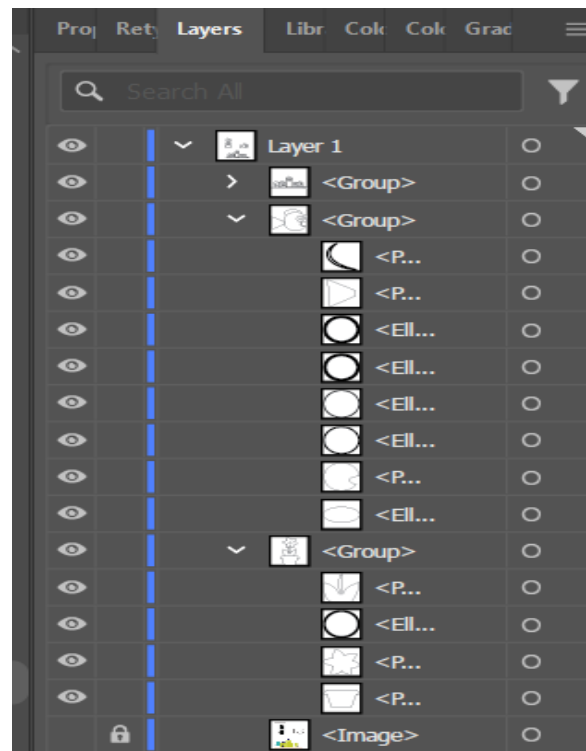
Adobe Illustrator

Experiment 1: Working with shapes Design a shape provided in lab class.

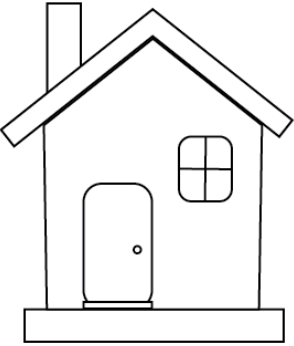
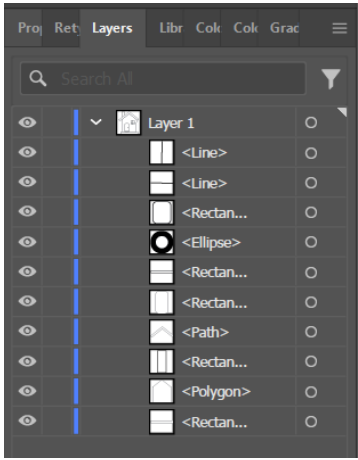
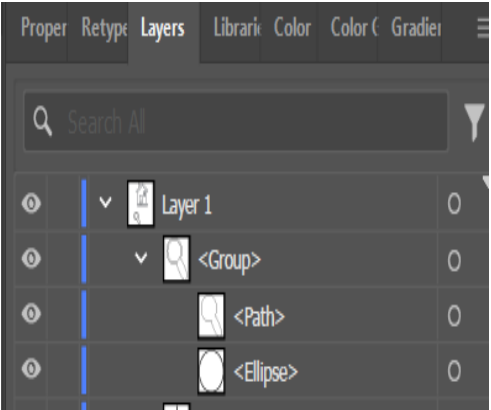
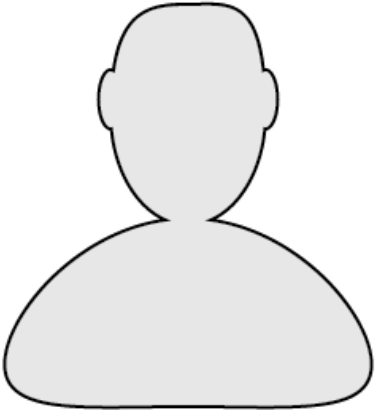
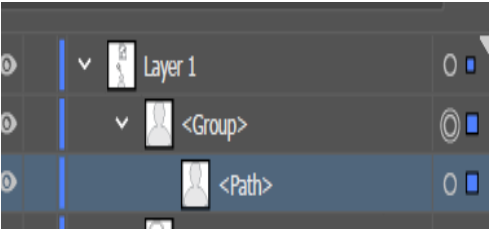
Design:

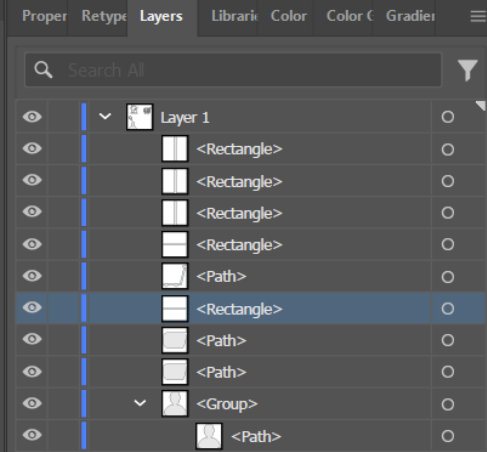
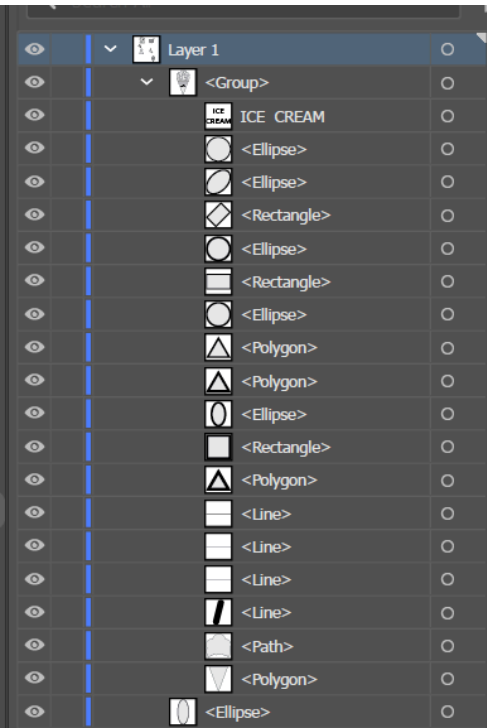


Layers used:



Experiment 2: Design basic icons Design 5 basic icons of your choice in illustrator.

Home icon		
Search icon		
User profile		

Cart icon		
Ice cream icon		

Experiment 3: Design a basic Flyer/Leaflets for information session.

IT INFORMATION SESSION

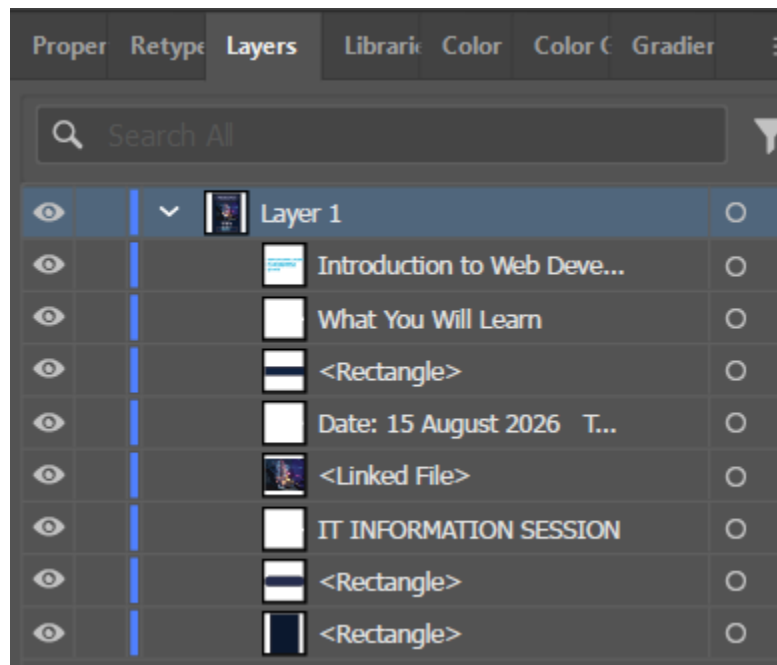


Date: 15 August 2026
Time: 11:00 AM – 2:00 PM
Location: Seminar Hall

What You Will Learn

- **Introduction to Web Development**
- **Career Opportunities in IT**
- **Live Project Demonstration**
- **Q&A Session**

Layers used



Lab 7: FFmpeg (Audio & Video Processing)

Objective: To learn how to process multimedia files using FFmpeg, including conversion, compression, editing, filtering, and optimization of audio and video.

Introduction: FFmpeg is a powerful open-source multimedia framework used for handling video, audio, and other multimedia files. It supports a wide range of operations such as format conversion, compression, trimming, filtering, and streaming.

Tools/Software Used:

- FFmpeg
- Command Line Interface (CMD / Terminal)

1. Basic Overview of FFmpeg

FFmpeg is a powerful open-source tool used for:

- Converting audio/video formats
- Editing, compressing, streaming
- Applying filters and effects

Basic syntax: `ffmpeg -i input.mp4 output.avi`

Video Tasks

1. Conversion

MP4 → AVI

```
ffmpeg -i input.mp4 output.avi
```

MP4 → MKV

```
ffmpeg -i input.mp4 output.mkv
```

Output

```
Microsoft Windows [Version 10.0.26200.8037]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 output.avi
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFmpeg developers
  built with gcc 15.2.0 (Crosstool-NG 1.28.0.23.185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threads --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enable-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disable-libxcb --disable-xlib --disable-libpulse --enable-openccl --enable-gmp --enable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libaribb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libdav1d --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ffnvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvazaar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libbxml --enable-libbzip3 --enable-libopus --enable-libplacebo --enable-librist --enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-libopav --enable-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enable-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --enable-schannel --enable-sdl2 --enable-libsnappy --enable-libsoxr --enable-libsrtp --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-libdrm --enable-vaapi --enable-libvidstab --enable-libvnc --enable-whisper --enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enable-libzimg --extra-libs=lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc=x86_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gcc-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale     9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
  frame= 17 fps=0.0 q=31.0 size= 256KiB time=00:00:01.24 bitrate=1679.1k
  frame= 77 fps= 74 q=31.0 size= 768KiB time=00:00:03.81 bitrate=1648.3k
  frame= 131 fps= 85 q=31.0 size= 1280KiB time=00:00:06.07 bitrate=1726.6k
  frame= 190 fps= 92 q=31.0 size= 1792KiB time=00:00:08.54 bitrate=1718.0k
  frame= 242 fps= 94 q=24.8 size= 2048KiB time=00:00:10.84 bitrate=1546.4k
[out#0/avi @ 0000025bb9dcb4c0] video:2571KiB audio:200KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: 1.064153%
  frame= 305 fps=101 q=31.0 Lsize= 2800KiB time=00:00:12.73 bitrate=1801.0 kbits/s speed= 4.2x elapsed=0:00:03.03

C:\Users\Dell\Pictures\Saved Pictures\video>
```

Pictures > Saved Pictures > video					Search video
Sort View ... Details					
Name	Date	Type	Size	Tag	
input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB		
output.avi	4/9/2026 3:00 PM	AVI File	2,800 KB		

2. Compression (reduce file size)

ffmpeg -i input.mp4 -vcodec libx264 -crf 28 output.mp4

- Lower CRF = better quality (18–28 recommended)

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -vcodec libx264 -crf 28 output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFmpeg developers
  built with gcc 15.2.0 (crossstool-NG 1.28.0.23.185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threads --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enable-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --enable-libxcb --disable-xlib --disable-libpulse --enable-openccl --enable-gmp --enable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libaribb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libdav1d --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ffnvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvazaar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libbxml --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist --enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-libao --enable-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enable-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --enable-schannel --enable-sdl2 --enable-lbsnappy --enable-libsoxr --enable-librt --enable-lbsvtav1 --enable-libtwolame --enable-libuavs3d --enable-libdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper --enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enable-libzimg --enable-libzvb --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxflags=-extra-libs=lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc=x86_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gcc-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale     9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
```

```
frame= 101 fps= 21 q=33.0 size=       768KiB time=00:00:07.46 bitrate= 842.7k
frame= 122 fps= 23 q=33.0 size=       1024KiB time=00:00:08.68 bitrate= 966.2k
frame= 141 fps= 24 q=33.0 size=       1024KiB time=00:00:09.47 bitrate= 895.7k
frame= 161 fps= 26 q=33.0 size=       1280KiB time=00:00:10.26 bitrate=1021.9k
frame= 177 fps= 26 q=33.0 size=       1280KiB time=00:00:11.05 bitrate= 948.9k
frame= 189 fps= 25 q=33.0 size=       1280KiB time=00:00:11.54 bitrate= 908.6k
frame= 205 fps= 26 q=33.0 size=       1536KiB time=00:00:12.20 bitrate=1031.2k
frame= 221 fps= 26 q=33.0 size=       1536KiB time=00:00:12.58 bitrate= 999.7k
frame= 227 fps= 25 q=33.0 size=       1536KiB time=00:00:09.38 bitrate=1340.9k
frame= 263 fps= 27 q=33.0 size=       2048KiB time=N/A bitrate=N/A speed=N/A e
frame= 287 fps= 28 q=1.0 size=       2048KiB time=N/A bitrate=N/A speed=N/A e
[Output#0/mp4 @ 000001760b0dc5c0] video:2417KiB audio:4KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.414037%
frame= 305 fps= 30 q=1.0 size=       2431KiB time=00:00:12.73 bitrate=1563.6
kb/s/s speed=1.24x elapsed=0:00:10.30
[Libx264 @ 000001760d67f040] frame I:3 Avg QP:22.18 size: 52934
[Libx264 @ 000001760d67f040] frame P:77 Avg QP:25.81 size: 19144
[Libx264 @ 000001760d67f040] frame B:225 Avg QP:28.38 size: 3741
[Libx264 @ 000001760d67f040] consecutive B-frames: 1.0% 1.3% 2.0% 95.7%
[Libx264 @ 000001760d67f040] mb I I16..4: 49.4% 42.6% 8.1%
[Libx264 @ 000001760d67f040] mb P I16..4: 4.4% 5.0% 0.2% P16..4: 42.6%
6.8% 3.6% 0.0% 0.0% skip:37.3%
[Libx264 @ 000001760d67f040] mb B I16..4: 0.2% 0.2% 0.0% B16..8: 32.2%
0.7% 0.1% direct: 0.2% skip:66.3% L0:37.9% L1:61.0% BI: 1.1%
[Libx264 @ 000001760d67f040] 8x8 transform intra:49.3% inter:82.5%
[Libx264 @ 000001760d67f040] coded y,uvDC,uvAC intra: 17.7% 53.0% 8.9% inter
: 3.1% 8.8% 0.4%
[Libx264 @ 000001760d67f040] i16 v,h,dc,p: 43% 23% 16% 18%
[Libx264 @ 000001760d67f040] i8 v,h,dc,ddl,ddr,vr,hd,vl,hu: 15% 13% 50% 3%
4% 5% 4% 3% 3%
[Libx264 @ 000001760d67f040] i4 v,h,dc,ddl,ddr,vr,hd,vl,hu: 23% 19% 15% 5%
9% 11% 7% 5% 5%
[Libx264 @ 000001760d67f040] i8c dc,h,v,p: 58% 20% 17% 5%
[Libx264 @ 000001760d67f040] Weighted P-Frames: Y:0.0% UV:0.0%
[Libx264 @ 000001760d67f040] ref P L0: 66.1% 12.2% 17.3% 4.5%
[Libx264 @ 000001760d67f040] ref B L0: 93.0% 5.8% 1.2%
[Libx264 @ 000001760d67f040] ref B L1: 97.1% 2.9%
[Libx264 @ 000001760d67f040] kb/s:1556.24
[aac @ 000001760d351500] Qavg: 65536.000
```

Pictures > Saved Pictures > video					
Search video					
Sort View Details					
Name	Date	Type	Size	T	
input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB		
output.avi	4/9/2026 3:00 PM	AVI File	2,800 KB		
output.mp4	4/9/2026 3:04 PM	MP4 File	2,431 KB		

3. Adjust Video Quality

Change bitrate

ffmpeg -i input.mp4 -b:v 1000k output.mp4

Change resolution

ffmpeg -i input.mp4 -vf scale=1280:720 output.mp4

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -b:v 1000k o
utput.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openssl --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkva
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enab
le-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrtp --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disab
le-lldrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzynthi --extra-flags=DLIBTWOLAME_STATIC --extra-cxxfla
gs=--extra-lib=lgomp --extra-ldflags=-pthread --extra-ldexeflags=-c-x86
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --
extra-version=20260407
  libavutil      60. 29.100 / 60. 29.100
  libavcodec     62. 29.101 / 62. 29.101
  libavformat    62. 13.101 / 62. 13.101
  libavdevice    62.  4.100 / 62.  4.100
  libavfilter    11. 15.101 / 11. 15.101
  libswscale     9.  7.100 /  9.  7.100
  libswresample  6.  4.100 /  6.  4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
  Metadata:
    major_brand      : mp42
    frame=      66 fps= 21 q=39.0 size=      256KiB time=00:00:06.61 bitrate= 317.2k
    frame=     86 fps= 23 q=38.0 size=      256KiB time=00:00:06.99 bitrate= 299.8k
    frame=    110 fps= 26 q=37.0 size=      512KiB time=00:00:08.23 bitrate= 509.4k
    frame=    133 fps= 28 q=37.0 size=      512KiB time=00:00:09.19 bitrate= 456.2k
    frame=    157 fps= 30 q=36.0 size=      512KiB time=00:00:10.60 bitrate= 395.6k
    frame=    181 fps= 31 q=36.0 size=      768KiB time=00:00:11.73 bitrate= 536.2k
    frame=    201 fps= 32 q=36.0 size=      768KiB time=00:00:12.24 bitrate= 513.8k
    frame=    220 fps= 32 q=36.0 size=      768KiB time=00:00:09.09 bitrate= 692.0k
    frame=    236 fps= 32 q=37.0 size=    1024KiB time=N/A bitrate=N/A speed=N/A e
    frame=    275 fps= 35 q=37.0 size=    1280KiB time=N/A bitrate=N/A speed=N/A e
[out#0/mp4 @ 0000018a37f0e6c0] video:1507KiB audio:4KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.663695%
frame=   305 fps= 37 q=-1.0 Lsize=    1520KiB time=00:00:12.73 bitrate= 977.9
kbits/s speed=1.56x elapsed=0:00:08.18
[libx264 @ 0000018a3a405040] frame I:3 Avg QP:25.51 size: 40495
[libx264 @ 0000018a3a405040] frame P:77 Avg QP:30.00 size: 11351
[libx264 @ 0000018a3a405040] frame B:225 Avg QP:33.58 size: 2430
[libx264 @ 0000018a3a405040] consecutive B-frames: 1.0% 1.3% 2.0% 95.7%
[libx264 @ 0000018a3a405040] mb I  I16..4: 54.2% 38.8% 7.0%
[libx264 @ 0000018a3a405040] mb P  I16..4: 3.3% 4.0% 0.1% P16..4: 32.9%
4.0% 1.9% 0.0% 0.0% skip:53.9%
[libx264 @ 0000018a3a405040] mb B  I16..4: 0.1% 0.2% 0.0% B16..8: 26.0%
0.3% 0.0% direct: 0.1% skip:73.4% L0:35.9% L1:63.7% BI: 0.4%
[libx264 @ 0000018a3a405040] final ratefactor: 31.45
[libx264 @ 0000018a3a405040] 8x8 transform intra:49.5% inter:85.3%
[libx264 @ 0000018a3a405040] coded y,uvDC,uvAC intra: 15.2% 43.7% 6.5% inter
: 1.5% 4.1% 0.2%
[libx264 @ 0000018a3a405040] i16 v,h,dc,p: 49% 22% 14% 16%
[libx264 @ 0000018a3a405040] i8 v,h,dc,ddl,ddr,vr,hd,vl,hu: 12% 10% 56% 3%
4% 5%
[libx264 @ 0000018a3a405040] i4 v,h,dc,ddl,ddr,vr,hd,vl,hu: 24% 19% 17% 5%
8% 11% 7% 5% 4%
[libx264 @ 0000018a3a405040] i8c dc,h,v,p: 65% 17% 14% 4%
[libx264 @ 0000018a3a405040] Weighted P-Frames: Y:0.0% UV:0.0%
[libx264 @ 0000018a3a405040] ref P L0: 64.3% 14.5% 17.4% 3.8%
[libx264 @ 0000018a3a405040] ref B L0: 94.0% 4.9% 1.1%
[libx264 @ 0000018a3a405040] ref B L1: 97.8% 2.2%
[libx264 @ 0000018a3a405040] kb/s:969.83
[aac @ 0000018a39aee440] Qavg: 65536.000
```

4. Trim Video

ffmpeg -i input.mp4 -ss 00:00:10 -to 00:00:30 -c copy output.mp4

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -ss 00:00:10
-to 00:00:30 -c copy output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crossstool-NG 1.28.0.23_185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkva
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enab
le-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsnappy --enable-libsoxr --enable-
librt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disabl
ibdcm --enable-vaapi --enable-libvidstab --enable-libvvec --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvb1 --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
extra-version=20260407
libavutil      60. 29.100 / 60. 29.100
libavcodec     62. 29.101 / 62. 29.101
libavformat    62. 13.101 / 62. 13.101
libavdevice    62.  4.100 / 62.  4.100
libavfilter    11. 15.101 / 11. 15.101
libswscale      9.  7.100 /  9.  7.100
libswresample  6.  4.100 /  6.  4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
  metadata_name      : E-SMASH Audio metadata
Press [q] to stop, [?] for help
[Out#0/mp4 @ 000002973e17ad80] video:535KiB audio:84KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.372356%
frame= 17 fps=0.0 q=-1.0 Lsize= 622KiB time=00:00:12.73 bitrate= 399.8
kbits/s speed= 415x elapsed=0:00:00.03

C:\Users\Dell\Pictures\Saved Pictures\video>
```

Pictures > Saved Pictures > video					Search video
Sort View ... Details					
Name	Date	Type	Size	Tag	
input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB		
output.mp4	4/9/2026 3:12 PM	MP4 File	622 KB		

5. Merge Videos

a) Using concat

Create file list.txt

file 'video1.mp4'

file 'video2.mp4'

ffmpeg -f concat -safe 0 -i list.txt -c copy output.mp4

Output

```
C:\Users\De\l\Pictures\Saved Pictures\video>ffmpeg -f concat -safe 0 -i list
txt -c copy output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (Crosstool-NG 1.28.0.23-185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkva
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable-
libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-libopav --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsndio --enable-libsrt --enable-
libsrtp --enable-libsvtav1 --enable-libtesseract --enable-libuv --enable-l
ibx264 --enable-libx265 --enable-libx265 --enable-libx265 --enable-libx265
--enable-libx265 --enable-libx265 --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs=-extra-libs=lgomp --extra-ldflags=-pthread --extra-ldexeflags=-ccx8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
extra-version=20260407
libavutil 60. 29.100 / 60. 29.100
libavcodec 62. 29.101 / 62. 29.101
libavformat 62. 13.101 / 62. 13.101
libavdevice 62. 4.100 / 62. 4.100
libavfilter 11. 15.101 / 11. 15.101
libswscale 9. 7.100 / 9. 7.100
libswresample 6. 4.100 / 6. 4.100
[mov,mp4,m4a,3gp,3g2,mj2 @ 0000025b52cdf740] Auto-inserting h264_mp4toannexb
bitstream filter
Press [q] to stop, [?] for help
[mov,mp4,m4a,3gp,3g2,mj2 @ 0000025b52d5a480] Auto-inserting h264_mp4toannexb
bitstream filter
[mov,mp4,m4a,3gp,3g2,mj2 @ 0000025b52cdf940] Auto-inserting h264_mp4toannexb
bitstream filter
[out#0/mp4 @ 0000025b52d4b740] video:24627KiB audio:1182KiB subtitle:0KiB ot
her streams:0KiB global headers:0KiB muxing overhead: 0.104736%
frames 915 fps=0.0 q=1.0 lsize= 25336KiB time=00:00:38.20 bitrate=5539.7
kbits/s speed= 163x elapsed=0:00:00.23
```

Name	Date	Type	Size
list.txt	4/9/2026 3:26 PM	Text Document	1 KB
output.mp4	4/9/2026 3:27 PM	MP4 File	25,837 KB
video1.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB
video2.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB
video3.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB

b) Using filter_complex

ffmpeg -i v1.mp4 -i v2.mp4 -filter_complex "[0:v][1:v]concat=n=2:v=1:a=0" output.mp4

Output

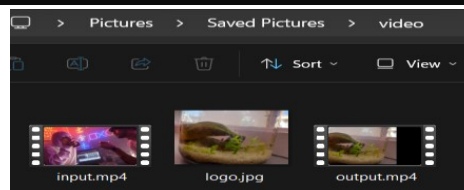
```
C:\Users\De\l\Pictures\Saved Pictures\video>ffmpeg -i v1.mp4 -i v2.mp4 -filt
er_complex "[0:v][1:v]concat=n=2:v=1:a=0" output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (Crosstool-NG 1.28.0.23-185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkva
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable-
libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-libopav --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsndio --enable-libsrt --enable-
libsrtp --enable-libsvtav1 --enable-libtesseract --enable-libuv --enable-l
ibx264 --enable-libx265 --enable-libx265 --enable-libx265 --enable-libx265
--enable-libx265 --enable-libx265 --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs=-extra-libs=lgomp --extra-ldflags=-pthread --extra-ldexeflags=-ccx8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
extra-version=20260407
libavutil 60. 29.100 / 60. 29.100
libavcodec 62. 29.101 / 62. 29.101
libavformat 62. 13.101 / 62. 13.101
libavdevice 62. 4.100 / 62. 4.100
libavfilter 11. 15.101 / 11. 15.101
libswscale 9. 7.100 / 9. 7.100
libswresample 6. 4.100 / 6. 4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'v1.mp4':
```



```

frame= 49 fps= 19 q=28.0 size= 256KiB time=00:00:06.10 bitrate= 343.8k
frame= 65 fps= 20 q=28.0 size= 256KiB time=00:00:06.61 bitrate= 317.2k
frame= 85 fps= 23 q=28.0 size= 512KiB time=00:00:07.63 bitrate= 549.2k
frame= 108 fps= 25 q=28.0 size= 512KiB time=00:00:08.14 bitrate= 514.7k
frame= 123 fps= 25 q=28.0 size= 768KiB time=00:00:09.17 bitrate= 685.9k
frame= 149 fps= 28 q=28.0 size= 768KiB time=00:00:10.19 bitrate= 617.0k
frame= 175 fps= 29 q=28.0 size= 1024KiB time=00:00:11.22 bitrate= 747.6k
frame= 200 fps= 30 q=28.0 size= 1280KiB time=00:00:12.24 bitrate= 856.3k
frame= 225 fps= 31 q=28.0 size= 1280KiB time=00:00:09.30 bitrate=1127.4k
frame= 253 fps= 33 q=28.0 size= 1792KiB time=N/A bitrate=N/A speed=N/A e
frame= 282 fps= 34 q=28.0 size= 1792KiB time=N/A bitrate=N/A speed=N/A e
[Out#0/mp4 @ 000001a5dcd4f2c0] video:2082KiB audio:4KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.477261%
frame= 305 fps= 36 q=-1.0 Lsize= 2095KiB time=00:00:12.73 bitrate=1347.8
kbits/s speed=1.49x elapsed=0:00:08.53
[libx264 @ 000001a5dcd4f2c0] frame I:2 Avg QP:16.02 size:149109
[libx264 @ 000001a5dcd4f2c0] frame P:79 Avg QP:20.63 size: 13352
[libx264 @ 000001a5dcd4f2c0] frame B:224 Avg QP:22.33 size: 3474
[libx264 @ 000001a5dcd4f2c0] consecutive B-frames: 0.7% 0.7% 10.8% 87.9%
[libx264 @ 000001a5dcd4f2c0] mb I I16..4: 14.4% 70.8% 14.8%
[libx264 @ 000001a5dcd4f2c0] mb P I16..4: 3.8% 5.2% 0.4% P16..4: 21.1%
4.4% 2.1% 0.0% 0.0% skip:63.1%
[libx264 @ 000001a5dcd4f2c0] mb B I16..4: 0.4% 0.4% 0.0% B16..8: 16.1%
0.8% 0.1% direct: 1.3% skip:80.8% L0:40.5% L1:57.1% BI: 2.4%
[libx264 @ 000001a5dcd4f2c0] 8x8 transform intra:57.4% inter:88.3%
[libx264 @ 000001a5dcd4f2c0] coded y,uvDC,uvAC intra: 30.1% 66.1% 21.5% inte
r: 2.6% 9.0% 0.1%
[libx264 @ 000001a5dcd4f2c0] i16 v,h,dc,p: 33% 26% 24% 17%
[libx264 @ 000001a5dcd4f2c0] i8 v,h,dc,ddl,ddr,vr,hd,vl,hu: 21% 17% 42% 3%
4% 3% 4% 3% 3%
[libx264 @ 000001a5dcd4f2c0] i4 v,h,dc,ddl,ddr,vr,hd,vl,hu: 28% 24% 19% 4%
7% 5% 7% 3% 3%
[libx264 @ 000001a5dcd4f2c0] i8c dc,h,v,p: 51% 23% 20% 6%
[libx264 @ 000001a5dcd4f2c0] Weighted P-Frames: Y:0.0% UV:0.0%
[libx264 @ 000001a5dcd4f2c0] ref P L0: 67.6% 7.6% 18.1% 6.7%
[libx264 @ 000001a5dcd4f2c0] ref B L0: 88.8% 9.0% 2.2%
[libx264 @ 000001a5dcd4f2c0] ref B L1: 95.4% 4.6%
[libx264 @ 000001a5dcd4f2c0] kb/s:1340.27
[aac @ 000001a5dcd4f2c0] Qavg: 65536.000

```



Name	Date	Type	Size	Length
input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	00:00:12
logo.jpg	12/8/2023 3:43 PM	JPG File	89 KB	
output.mp4	4/9/2026 3:37 PM	MP4 File	2,096 KB	00:00:12

7. Filters

Greyscale

```
ffmpeg -i input.mp4 -vf format=gray output.mp4
```

Brightness/contrast

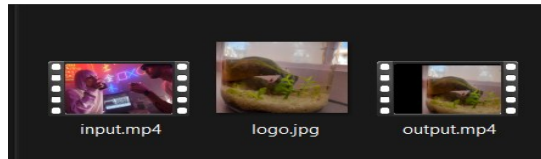
```
ffmpeg -i input.mp4 -vf eq=brightness=0.05:contrast=1.5 output.mp4
```

Watermark

```
ffmpeg -i input.mp4 -i logo.png -filter_complex "overlay=W-w-10:H-h-10" output.mp4
```


Output

```
C:\Users\ DELL \Pictures\Saved Pictures\Videos>ffmpeg -i input.mp4 -i logo.jpg
-filter_complex "overlay=W-w-10:H-h-10" output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 The FFm
peg developers
built with gcc 15.2.0 (Crosstool-NG 1.28.0-pkg-config348)
configuration: --prefix=ffbuild/prefix --pkg-config-flags=--static --pkgco
nfig-checker=x86_64-w64-mingw32 --arch=x86_64 --target-os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32thre
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxlm2 --enabl
e-libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-lbfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --enable-xlib --disable-libpulse --enable-openc1 --enable-gmp --enab
le-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
mv --enable-avutil --enable-swscaler --enable-ladspa --enable-lavfi --enable-lbda
vs2 --enable-libdavdread --enable-libdvnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-lbgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvp8 --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-libopavp --enab
le-libopenhore --enable-libopenhore --enable-libopenh264 --enable-libopenhe
re --enable-libopenjpeg --enable-libomp --enable-librav1e --enable-librubberband
--enable-schannel --enable-sdl2 --enable-libsnpappy --enable-libsoxr --enable
-libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvnc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-lbzimg --enable-libzbvi --extra-cflags=-DLIBXAVS2_STATIC --extra-cxxfla
gs=-extra-libs=loomp extra-ldflags=-pthread --extra-ldexeflags=-ccs=x8
6_64-w64-mingw32-gcc-cross-prefix=x86_64-w64-mingw32-g++-cross-prefix=x86_64-w64-mingw32-gc
-car -ranlib=x86_64-w64-mingw32-gcc-ranlib -nm=x86_64-w64-mingw32-gcc-nm
-extra-version=20260407
lavutil 60. 29.100 / 60. 29.100
lavacodec 62. 29.101 / 62. 29.101
lavformat 62. 13.101 / 62. 13.101
lavdevice 62. 4.100 / 62. 4.100
lavfilter 11. 15.101 / 11. 15.101
lavrescale 9. 7.100 / 9. 7.100
libswresample 6. 4.100 / 6. 4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
Metadata:
major_brand : mp42
```



A screenshot of a Windows File Explorer window. The address bar shows the path: Pictures > Saved Pictures > video. The search bar contains the text 'Search video'. The ribbon at the top has icons for 'Home', 'Share', 'Move to', 'Delete', 'Sort', 'View', and 'Details'. The 'View' dropdown is set to 'View'. The 'Details' pane is open on the right. The main area shows a list of files with columns: Name, Date, Type, Size, and Length. The files listed are: 'input.mp4' (2/27/2021 3:16 PM, MP4 File, 8,611 KB, 00:00:12), 'logo.jpg' (12/8/2023 3:43 PM, JPG File, 89 KB, 00:00:00), and 'output.mp4' (4/9/2026 3:40 PM, MP4 File, 2,522 KB, 00:00:12).

8. Extract Audio / Video

```
# Extract audio
```

```
ffmpeg -i input.mp4 -q:a 0 -map a output.mp3
```

```
# Remove audio (video only)
```

```
ffmpeg -i input.mp4 -an output.mp4
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -q:a 0 -map
a output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crossstool-NG 1.28.0.23.185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-lbsnappy --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disabl
e-libdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs=-extra-libs=lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
extra-version=20260407
libavutil      60. 29.100 / 60. 29.100
libavcodec     62. 29.101 / 62. 29.101
libavformat    62. 13.101 / 62. 13.101
libavdevice    62.  4.100 / 62.  4.100
libavfilter    11. 15.101 / 11. 15.101
libswscale     9.  7.100 /  9.  7.100
libswresample  6.  4.100 /  6.  4.100
Input #0: mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
  duration: 00:00:12.73, start: 0.000000, bitrate: 32.3kbts/s speed=97.5x elapsed=
0:00:00.13
[Out#0/mp3 @ 0000027e05b88c00] video:0KiB audio:50KiB subtitle:0KiB other st
reams:0KiB global headers:0KiB muxing overhead: 0.661811%
size= 50KiB time=00:00:12.73 bitrate= 32.3kbts/s speed=97.5x elapsed=
0:00:00.13

C:\Users\Dell\Pictures\Saved Pictures\video>
```

File Name	Created	File Type	Size	Duration
input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	00:00:12
output.mp3	4/9/2026 3:44 PM	MP3 File	51 KB	00:00:12

9. Combine Audio & Video

ffmpeg -i video.mp4 -i audio.mp3 -c:v copy -c:a aac output.mp4

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -i audio.mp3
-map 0:v -map 1:a -c:v copy -c:a aac output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crossstool-NG 1.28.0.23.185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-lbsnappy --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disabl
e-libdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs=-extra-libs=lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
extra-version=20260407
libavutil      60. 29.100 / 60. 29.100
libavcodec     62. 29.101 / 62. 29.101
libavformat    62. 13.101 / 62. 13.101
libavdevice    62.  4.100 / 62.  4.100
libavfilter    11. 15.101 / 11. 15.101
libswscale     9.  7.100 /  9.  7.100
libswresample  6.  4.100 /  6.  4.100
```

```
C:\Users\ DELL\Pictures\Saved Pictures>Video>ffmpeg -framerate 1/3 -i img%0.jpg  
pg -c:v libx264 -r 30 -goutint.mp4  
ffmpeg version N-123858-g90510aff11b-20260407 Copyright (c) 2000-2026 the FFmpeg developers  
built with gcc 15.2.0 (crossstool-NG 1.28.0.23_185f348)  
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-config-pkg-config --cross-prefix=x86_64-w64-mingw32 --arch=x86_64 --target-os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threads --enable-pthreads --enable-iconv --enable-zlib --enable-lzma --enable-libxml2 --enable-libmva --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enable-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disable-libxcb --disable-xlib --enable-libpulse --enable-opengl --enable-gmp --enable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libaribb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libdav1d --enable-libddvdread --enable-libdvdrnav --disable-libfdk-aac --enable-ffnvcodec --enable-cuda-llvm --enable-freeio --enable-libgme --enable-libkvazaar --enable-libraricaaption --enable-libass --enable-libbluray --enable-libjxl --enable-libbmp3lame --enable-libopus --enable-libplacebo --enable-librist --enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-libopenaptv --enable-libopencore-amrn --enable-libopencore-amrw --enable-libopenh264 --enable-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --enable-channel --enable-sdl2 --enable-lisnappy --enable-libsoxr --enable-librt --enable-libstvapi --enable-libtwtolame --enable-libuavs3d --disable-libdrm --enable-vaapi --enable-libvidstab --enable-libvnc --enable-whisper --enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enable-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxflags=-extra-libs=lgomp --extra-ldflags=pthread --extra-ldexeflags=-cx=x86_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gcc-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --extra-version=20260407  
  
libavutil 60. 29.100 / 60. 29.100  
libavcodec 62. 29.101 / 62. 29.101  
libavformat 62. 13.101 / 62. 13.101  
libavdevice 62. 4.100 / 62. 4.100  
libavfilter 11. 15.101 / 11. 15.101  
libbswscale 9. 7.100 / 9. 7.100  
libswresample 6. 4.100 / 6. 4.100
```

```
Input #00, image2, from '':  
frame= 310 fps=9.3 q=29.0 size= 4352KiB time=00:00:10.40 bitrate=3400.2K  
frame= 324 fps=5.4 q=29.0 size= 4352KiB time=00:00:10.73 bitrate=3321.6K  
frame= 340 fps=5.6 q=1.0 size= 4352KiB time=00:00:11.26 bitrate=3164.4K  
frame= 356 fps=5.8 q=1.0 size= 4352KiB time=00:00:11.40 bitrate=3127.4K  
[Out#0/mp4 @ 000001c322ca9340] video:4456KiB audio:0KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: 0.183151%  
frame= 360 fps=5.8 q=1.0 Lsize= 4458KiB time=00:00:11.93 bitrate=3060.5 kbits/s dup=356 drop=0 speed=0.194x elapsed=0:01:01.54  
[Libx264 @ 000001c324c87ac0] frame I:4 Avg QP:15.00 size:1095588  
[Libx264 @ 000001c324c87ac0] frame P:92 Avg QP:18.13 size: 709  
[Libx264 @ 000001c324c87ac0] frame B:264 Avg QP:22.86 size: 412  
[Libx264 @ 000001c324c87ac0] consecutive B-frames: 2.2% 0.0% 97.8%  
[Libx264 @ 000001c324c87ac0] mb I 16..4 : 3.7% 84.8% 11.5%  
[Libx264 @ 000001c324c87ac0] mb P 16..4 : 0.0% 0.0% 0.0% P16..4 : 0.5% 0.0% 0.0% 0.0% skip:99.5%  
[Libx264 @ 000001c324c87ac0] mb B 16..4 : 0.0% 0.0% 0.0% B16..8 : 0.2% 0.0% 0.0% direct: 0.0% skip:99.8% L0:46.0% L1:54.0% BI: 0.0%  
[Libx264 @ 000001c324c87ac0] 8x8 transform intra:84.8% inter:95.9%  
[Libx264 @ 000001c324c87ac0] coded y,u,v,c intra: 87.7% 64.0% 39.2% inte r= 0.0% 1% 0.0%  
[Libx264 @ 000001c324c87ac0] i16 v,h,d,c,p: 39% 21% 28% 11%  
[Libx264 @ 000001c324c87ac0] i8 v,h,d,c,ddl,ldr,vr,hd,vl,hu: 23% 21% 28% 4% 4% 5% 4% 5% 6%  
[Libx264 @ 000001c324c87ac0] i4 v,h,d,c,ddl,ddr,vr,hd,vl,hu: 46% 17% 6% 4% 6% 7% 4% 6% 3%  
[Libx264 @ 000001c324c87ac0] i8c dc,h,v,p: 39% 23% 31% 7%  
[Libx264 @ 000001c324c87ac0] Weighted P-Frames: Y:0.0% UV:0.0%  
[Libx264 @ 000001c324c87ac0] ref p L0: 32.6% 0.1% 2.7% 3.7%  
[Libx264 @ 000001c324c87ac0] ref B L0: 8.6% 6.0% 7.1%  
[Libx264 @ 000001c324c87ac0] ref B L1: 99.1% 0.9%  
[Libx264 @ 000001c324c87ac0] kb/s:3037.47
```


12. Create GIF

```
ffmpeg -i input.mp4 -vf "fps=10,scale=320:-1" output.gif
```

Output

```
C:\Users\ DELL\Pictures\Saved Pictures>video ffmpeg -i input.mp4 -vf "fps=10,scale=320:-1" output.gif  
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFmpeg developers  
built with gcc 15.2.0 (CrossTool-NG 1.28.0.23.185f348)  
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-config=pkg-config --cross-prefix=x86_64-w64-mingw32 --arch=x86_64 --target-os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threads --enable-pthread --enable-iconv --enable-zlib --enable-lxml2 --enable-libvmafa --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enable-libribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disable-libxcb --disable-xlib --disable-lbpulse --enable-opengl --enable-gmp --enable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libaribb24 --enable-avisynth --enable-chromaprint --enable-libdavld --enable-libdavs2 --enable-libdvdxread --enable-libdvndav --disable-libfdk-aac --enable-ffnvcodec --enable-cuda-llvm --enable-freeioR --enable-libgme --enable-libkvazaar --enable-liblavcapturen --enable-libass --enable-libbluray --enable-libbixbyl --enable-libbpm3lame --enable-libopus --enable-libplacebo --enable-librist --enable-libssh --enable-libtheora --enable-libvp9x --enable-libwebp --enable-libzimq --enable-lv2 --enable-libvpl --enable-openal --enable-libopavp --enable-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enable-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --enable-schannel --enable-sdl2 --enable-libsnappy --enable-libsoxr --enable-libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-libidrm --enable-vaapi --enable-libvidstab --enable-libvvenc --enable-whisper --enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enable-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-ccxfllags=-extra-libs=gomp --extra-ldflags=pthread --extra-ldexeflags=-cxcsx86_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ax=x86_64-w64-mingw32-gcc-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --extra-version=20260407  


|               |     |          |     |        |
|---------------|-----|----------|-----|--------|
| libavutil     | 60. | 29.100 / | 60. | 29.100 |
| libavcodec    | 62. | 29.101 / | 62. | 29.101 |
| libavformat   | 62. | 13.101 / | 62. | 13.101 |
| libavdevice   | 62. | 4.100 /  | 62. | 4.100  |
| libavfilter   | 11. | 15.101 / | 11. | 15.101 |
| libswscale    | 9.  | 7.100 /  | 9.  | 7.100  |
| libswresample | 6.  | 4.100 /  | 6.  | 4.100  |

  
frames= 29 fps=0.0 q=0.0 size= 256KiB time=00:00:02.90 bitrate= 723.2Kbps frame_size= 76 q=0.0 size= 1024KiB time=00:00:07.80 bitrate=1075.5Kbps frames= 123 fps= 79 q=0.0 size= 1792KiB time=00:00:12.30 bitrate=1193.5Kbps [out#0/gif @ 000001b58be4c1c0] video:1986KiB audio:0KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: 0.000984%  
frame_size= 127 fps= 80 q=9.0 L-size= 1586KiB time=00:00:12.70 bitrate=1280.8kbts/s speed=7.99x Elapsed=0:00:01.58
```

 input.mp4	2/27/2021 3:16 PM	MP4 File	8,611 KB	00:00:12
 output.gif	4/9/2026 4:27 PM	GIF File	1,986 KB	

13. Extract Frames

```
ffmpeg -i input.mp4 frame_%04d.png
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp4 frame_%04d.p
ng
ffmpeg version N-123858-g0510a6ff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crossstool-NG 1.28.0.23.185f348)
configuration: --prefix=ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
d --enable-threads --enable-iconv --enable-zlib --enable-libxslt --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-lbfridibi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-libcvcdec --enable-amf --enable-libaom --enable-libar
ibv24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkva
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libb
x --enable-libbpl --enable-libbsb --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-opengl --enable-libopenv --enab
le-libopencl --enable-amrnb --enable-libopencl --enable-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband -
--enable-schannel --enable-sdl2 --enable-lbsnappy --enable-libsoxr --enable-
libsr --enable-lbssvtav1 --enable-libtwolame --enable-libuavs3d --disabl
e-libdrrm --enable-vaapi --enable-libvidstab --enable-libvvc --enable-whisper
--enable-libz4 --enable-libz455 --enable-libxavs2 --enable-libxvid --enab
e-libzimg --enable-libzvtbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs=-extra-libs-lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc-x86
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
extra-version=20260407
libavutil 60. 29.100 / 60. 29.100
libavcodec 62. 29.101 / 62. 29.101
libavformat 62. 13.101 / 62. 13.101
libavdevice 62. 4.100 / 62. 4.100
libavfilter 11. 15.101 / 11. 15.101
libbswscale 9. 7.100 / 9. 7.100
libswresample 6. 4.100 / 6. 4.100
```

```

frame= 252 fps= 11 q=-0.0 size=N/A time=00:00:10.51 bitrate=N/A speed=0.479
frame= 260 fps= 12 q=-0.0 size=N/A time=00:00:10.84 bitrate=N/A speed=0.483
frame= 269 fps= 12 q=-0.0 size=N/A time=00:00:11.21 bitrate=N/A speed=0.487
frame= 277 fps= 12 q=-0.0 size=N/A time=00:00:11.55 bitrate=N/A speed=0.491
frame= 281 fps= 12 q=-0.0 size=N/A time=00:00:11.72 bitrate=N/A speed=0.486
frame= 291 fps= 12 q=-0.0 size=N/A time=00:00:12.13 bitrate=N/A speed=0.492
frame= 298 fps= 12 q=-0.0 size=N/A time=00:00:12.42 bitrate=N/A speed=0.493
[out#0/image2 @ 000002bc8f8f9400] video:483808KiB audio:0KiB subtitle:0KiB o
ther streams:0KiB global headers:0KiB muxing overhead: unknown
frame= 305 fps= 12 q=-0.0 Lsize=N/A time=00:00:12.72 bitrate=N/A speed= 0.5
x elapsed=0:00:25.46

```

Name	Date	Type	Size
 frame_0001.png	4/9/2026 4:30 PM	PNG File	9 KB
 frame_0002.png	4/9/2026 4:30 PM	PNG File	1,379 KB
 frame_0003.png	4/9/2026 4:30 PM	PNG File	1,487 KB
 frame_0004.png	4/9/2026 4:30 PM	PNG File	1,504 KB
 frame_0005.png	4/9/2026 4:30 PM	PNG File	1,516 KB
 frame_0006.png	4/9/2026 4:30 PM	PNG File	1,535 KB
 frame_0007.png	4/9/2026 4:30 PM	PNG File	1,544 KB

14. Slow Motion

ffmpeg -i input.mp4 -filter:v "setpts=2.0*PTS" output.mp4

Output

```

C:\Users\ DELL\ Pictures\Saved Pictures\video>ffmpeg -i input.mp4 -filter:v "s
etpts=2.0*PTS" output.mp4
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crossstool-NG 1.28.0.23_185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-ltpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-lbcevecdec --enable-amf --enable-libaom --enable-libar
ibp24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkva
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libs5h --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencl --enable-libopencl --enable-libopencl --enable-libopencl --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsnap --enable-libsoxr --enable-
libsr --enable-libsrt --enable-libswscale --enable-libtesseract --enable-l
ibdr --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzmbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs=-extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
extra-version=20260407
libavutil 60. 29.100 / 60. 29.100
libavcodec 62. 29.101 / 62. 29.101
libavformat 62. 13.101 / 62. 13.101
libavdevice 62. 4.100 / 62. 4.100
libavfilter 11. 15.101 / 11. 15.101
libswscale 9. 7.100 / 9. 7.100
libswresample 6. 4.100 / 6. 4.100
Input #0, mov,mp4,m4a,3gp,3g2,mj2, from 'input.mp4':
Metadata:
major_brand : mp42
frame= 227 fps= 26 q=26.0 size= 3584KiB time=00:00:18.72 bitrate=1567.8K
frame= 252 fps= 27 q=26.0 size= 3584KiB time=N/A bitrate=N/A speed=N/A e
frame= 272 fps= 28 q=26.0 size= 3584KiB time=N/A bitrate=N/A speed=N/A e
frame= 287 fps= 28 q=-1.0 size= 3840KiB time=N/A bitrate=N/A speed=N/A e
[out#0/mp4 @ 000002615fa594c0] video:7050KiB audio:4KiB subtitle:0KiB other
streams:0KiB global headers:0KiB muxing overhead: 0.126508%
frame= 305 fps= 29 q=-1.0 Lsize= 7062KiB time=00:00:25.23 bitrate=2292.7
kbits/s speed=2.37x elapsed=0:00:10.64
[libx264 @ 0000026161eb5400] frame I:3 Avg QP:14.94 size:105262
[libx264 @ 0000026161eb5400] frame P:77 Avg QP:18.68 size: 54281
[libx264 @ 0000026161eb5400] frame B:225 Avg QP:21.41 size: 12101
[libx264 @ 0000026161eb5400] consecutive B-frames: 1.0% 1.3% 2.0% 95.7%
[libx264 @ 0000026161eb5400] mb I I16..4: 43.0% 43.7% 13.3%
[libx264 @ 0000026161eb5400] mb P I16..4: 6.5% 9.1% 1.1% P16..4: 47.9%
18.8% 8.9% 0.0% 0.0% skip: 7.8%
[libx264 @ 0000026161eb5400] mb B I16..4: 1.2% 1.1% 0.0% B16..8: 45.9%
3.7% 0.4% direct: 3.7% skip:43.9% L0:42.4% L1:51.8% BI: 5.8%
[libx264 @ 0000026161eb5400] 8x8 transform intra:51.2% inter:76.0%
[libx264 @ 0000026161eb5400] coded y,uvDC,uvAC intra: 29.6% 71.8% 24.4% inte
r: 11.5% 29.5% 1.7%
[libx264 @ 0000026161eb5400] i16 v,h,dc,p: 36% 22% 24% 17%
[libx264 @ 0000026161eb5400] i8 v,h,dc,ddl,ddr,vr,hd,vl,hu: 20% 17% 42% 3%
4% 4% 4% 3% 3%
[libx264 @ 0000026161eb5400] i4 v,h,dc,ddl,ddr,vr,hd,vl,hu: 26% 21% 22% 4%
7% 8% 6% 4% 4%
[libx264 @ 0000026161eb5400] i8c dc,h,v,p: 51% 22% 19% 9%
[libx264 @ 0000026161eb5400] Weighted P-Frames: Y:0.0% UV:0.0%
[libx264 @ 0000026161eb5400] ref P L0: 63.4% 10.1% 19.8% 6.8%
[libx264 @ 0000026161eb5400] ref B L0: 87.8% 10.2% 2.1%
[libx264 @ 0000026161eb5400] ref B L1: 94.9% 5.1%
[libx264 @ 0000026161eb5400] kb/s:2269.68
[aac @ 0000026161b524c0] Oavg: 65536.000

```


2. Adjust Audio Quality

Bitrate

ffmpeg -i input.mp3 -b:a 192k output.mp3

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -b:a 192k ou
tput.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (Crosstool-NG 1.28.0.23-185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
r --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-libao --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsnappy --enable-libsoxr --enable-
libsrt --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --
extra-version=20260407
libavutil      60. 29.100 / 60. 29.100
libavcodec     62. 29.101 / 62. 29.101
libavformat    62. 13.101 / 62. 13.101
libavdevice    62.  4.100 / 62.  4.100
libavfilter    11. 15.101 / 11. 15.101
libswscale     9.  7.100 /  9.  7.100
libswresample  6.  4.100 /  6.  4.100
encoder: Lavc62.29.101 libmp3lame
[Out#0/mp3 @ 0000016ad41acc00] video:0KiB audio:299KiB subtitle:0KiB other s
treams:0KiB global headers:0KiB muxing overhead: 0.235615%
size=      300KiB time=00:00:12.73 bitrate= 192.9kbits/s speed=74.4x elapsed=
0:00:00.17
C:\Users\Dell\Pictures\Saved Pictures\video>|
```

 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 8:53 PM	MP3 File	300 KB	00:00:12

Sample rate

ffmpeg -i input.mp3 -ar 44100 output.mp3

Output

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:03 PM	MP3 File	200 KB	00:00:13

```
encoder: Lavc62.29.101 libmp3lame
[Out#0/mp3 @ 000001818d83adc0] video:0KiB audio:200KiB subtitle:0KiB other s
treams:0KiB global headers:0KiB muxing overhead: 0.173205%
size=      200KiB time=00:00:12.73 bitrate= 128.6kbits/s speed=80.3x elapsed=
0:00:00.15
```



```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -ar 44100 ou
tput.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsndpy --enable-libsoxr --enable-
libsrtp --enable-libsrt --enable-libtesseract --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvvc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
-extra-version=20260407
```

3. Audio Filters

Normalize

ffmpeg -i input.mp3 -af loudnorm output.mp3

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -af loudnorm
output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvaza
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsndpy --enable-libsoxr --enable-
libsrtp --enable-libsrt --enable-libtesseract --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvvc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
-extra-version=20260407
```

```
size=      57KiB time=00:00:03.60 bitrate= 128.7kb/s speed=    7x elapsed=
size=     132KiB time=00:00:08.40 bitrate= 128.3kb/s speed=8.13x elapsed=
[Out#0/mp3 @ 00000234f8e0d700] video:0KiB audio:200KiB subtitle:0KiB other s
trems:0KiB global headers:0KiB muxing overhead: 0.165453%
size=     200KiB time=00:00:12.80 bitrate= 127.9kb/s speed=9.09x elapsed=
0:00:01.40
```

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:11 PM	MP3 File	200 KB	00:00:13

Volume increase

```
ffmpeg -i input.mp3 -af "volume=2.0" output.mp3
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -af "volume=
2.0" output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (Crosstool-NG 1.28.0.23.185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkva
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrtp --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disabl
e-libdrm --enable-vaapi --enable-libvidstab --enable-libvnc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
--extra-version=20260407
encoder      : Lavc62.29.101 libmp3lame
[Out#0/mp3 @ 000001f9668eb800] video:0KiB audio:200KiB subtitle:0KiB other s
treams:0KiB global headers:0KiB muxing overhead: 0.165453%
size=       200KiB time=00:00:12.73 bitrate= 128.5kbits/s speed= 101x elapsed=
0:00:00.12
C:\Users\Dell\Pictures\Saved Pictures\video>
```

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:17 PM	MP3 File	200 KB	00:00:13

High-pass filter

```
ffmpeg -i input.mp3 -af "highpass=f=200" output.mp3
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -af "highpas
s=f=200" output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
  built with gcc 15.2.0 (Crosstool-NG 1.28.0.23.185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkva
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrtp --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disabl
e-libdrm --enable-vaapi --enable-libvidstab --enable-libvnc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
--extra-version=20260407
```

```
encoder      : Lavc62.29.101 libmp3lame
[out#0/mp3 @ 000001db4d9ad000] video:0KiB audio:200KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: 0.165453%
size=      200KiB time=00:00:12.73 bitrate= 128.5kbits/s speed= 111x elapsed= 0:00:00.11

C:\Users\Dell\Pictures\Saved Pictures\video>
```

Name	Date	Type	Size	Length
input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
output.mp3	4/9/2026 9:20 PM	MP3 File	200 KB	00:00:13

Low-pass filter

ffmpeg -i input.mp3 -af "lowpass=f=3000" output.mp3

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video> ffmpeg -i input.mp3 -af "lowpass=f=3000" output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFmpeg developers
  built with gcc 15.2.0 (cross-tool-NG 1.28.0.23.185f348)
  configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threads --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enable-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disable-libxcb --disable-xlib --disable-libpulse --enable-opengl --enable-gmp --enable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libaribb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libdav1d --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ffnvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkvazaar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjack --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist --enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-libopenal --enable-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --enable-libschannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-libsrtp --enable-libsvtav1 --enable-libtesseract --enable-libuavs3d --enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enable-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxflags= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags= --cc=x86_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gcc-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --extra-version=20260407

encoder      : Lavc62.29.101 libmp3lame
[out#0/mp3 @ 000002565221fbc0] video:0KiB audio:200KiB subtitle:0KiB other streams:0KiB global headers:0KiB muxing overhead: 0.165453%
size=      200KiB time=00:00:12.73 bitrate= 128.5kbits/s speed= 104x elapsed= 0:00:00.12

C:\Users\Dell\Pictures\Saved Pictures\video>
```

Name	Date	Type	Size	Length
input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
output.mp3	4/9/2026 9:25 PM	MP3 File	200 KB	00:00:13

Reverb

ffmpeg -i input.mp3 -af "aecho=0.8:0.9:1000:0.3" output.mp3

Output

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:26 PM	MP3 File	216 KB	00:00:14

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -af "aecho=0
.8:0.9:1000:0.3" output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crossstool-NG 1.28.0.23.185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkva
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencl --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrtp --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disabl
e-libvorbis --enable-libvpx --enable-libvidstab --enable-libvnc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvtbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
--extra-version=20260407
libavutil 60. 29.100 / 60. 29.100
encoder : Lavc62.29.101 libmp3lame
[out#0/mp3 @ 0000015893f5a440] video:0KiB audio:215KiB subtitle:0KiB other s
treams:0KiB global headers:0KiB muxing overhead: 0.153346%
size= 216KiB time=00:00:13.73 bitrate= 128.6kbits/s speed= 109x elapsed=
0:00:00.12

C:\Users\Dell\Pictures\Saved Pictures\video>
```

Pitch change

ffmpeg -i input.mp3 -af "asetrate=44100*1.25" output.mp3

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -af "asetrat
e=44100*1.25" output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crossstool-NG 1.28.0.23.185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkva
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencl --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrtp --enable-libsvtav1 --enable-libtwolame --enable-libuavs3d --disabl
e-libvorbis --enable-libvpx --enable-libvidstab --enable-libvnc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvtbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc=x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm -
--extra-version=20260407
```

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:28 PM	MP3 File	175 KB	00:00:11

[illegible]

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:29 PM	MP3 File	216 KB	00:00:14

```
ffmpeg -i input.mp3 -af "afftdn" output.mp3
```

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>ffmpeg -i input.mp3 -af "afftdn"
output.mp3
ffmpeg version N-123858-g0510aff11b-20260407 Copyright (c) 2000-2026 the FFm
peg developers
built with gcc 15.2.0 (crosstool-NG 1.28.0.23_185f348)
configuration: --prefix=/ffbuild/prefix --pkg-config-flags=--static --pkg-
config=pkg-config --cross-prefix=x86_64-w64-mingw32- --arch=x86_64 --target-
os=mingw32 --enable-gpl --enable-version3 --disable-debug --disable-w32threa
ds --enable-pthreads --enable-iconv --enable-zlib --enable-libxml2 --enable-
libvmaf --enable-fontconfig --enable-libharfbuzz --enable-libfreetype --enab
le-libfribidi --enable-vulkan --enable-libshaderc --enable-libvorbis --disab
le-libxcb --disable-xlib --disable-libpulse --enable-openc1 --enable-gmp --e
nable-lzma --enable-liblcevc-dec --enable-amf --enable-libaom --enable-libar
ibb24 --enable-avisynth --enable-chromaprint --enable-libdav1d --enable-libd
avs2 --enable-libdvdread --enable-libdvdnav --disable-libfdk-aac --enable-ff
nvcodec --enable-cuda-llvm --enable-frei0r --enable-libgme --enable-libkva
ar --enable-libaribcaption --enable-libass --enable-libbluray --enable-libjx
l --enable-libmp3lame --enable-libopus --enable-libplacebo --enable-librist
--enable-libssh --enable-libtheora --enable-libvpx --enable-libwebp --enable
-libzmq --enable-lv2 --enable-libvpl --enable-openal --enable-liboapv --enab
le-libopencore-amrnb --enable-libopencore-amrwb --enable-libopenh264 --enabl
e-libopenjpeg --enable-libopenmpt --enable-librav1e --enable-librubberband --
enable-schannel --enable-sdl2 --enable-libsndio --enable-libsoxr --enable-
libsrtp --enable-libsrt --enable-libtwolame --enable-libuavs3d --disable-l
ibdrm --enable-vaapi --enable-libvidstab --enable-libvenc --enable-whisper
--enable-libx264 --enable-libx265 --enable-libxavs2 --enable-libxvid --enabl
e-libzimg --enable-libzvbi --extra-cflags=-DLIBTWOLAME_STATIC --extra-cxxfla
gs= --extra-libs=-lgomp --extra-ldflags=-pthread --extra-ldexeflags=-cc-x8
6_64-w64-mingw32-gcc --cxx=x86_64-w64-mingw32-g++ --ar=x86_64-w64-mingw32-gc
c-ar --ranlib=x86_64-w64-mingw32-gcc-ranlib --nm=x86_64-w64-mingw32-gcc-nm --
extra-version=20260407
libavutil 60. 29.100 / 60. 29.100

encoder      : Lavc62.29.101 libmp3lame
[out#0/mp3 @ 000001d4ac61e440] video:0KiB audio:200KiB subtitle:0KiB other s
treams:0KiB global headers:0KiB muxing overhead: 0.165453%
size=      200KiB time=00:00:12.73 bitrate= 128.5kbits/s speed= 101x elapsed=
0:00:00.12

C:\Users\Dell\Pictures\Saved Pictures\video>
```

Name	Date	Type	Size	Length
 input.mp3	4/9/2026 8:47 PM	MP3 File	51 KB	00:00:12
 output.mp3	4/9/2026 9:31 PM	MP3 File	200 KB	00:00:13

Image Compression using cwebp

1. Basic Overview of cwebp

cwebp converts images to **WebP format**, which:

- Reduces size significantly
- Maintains quality
- Improves web performance

Tasks

a) Image Conversion

cwebp input.jpg -o output1.webp

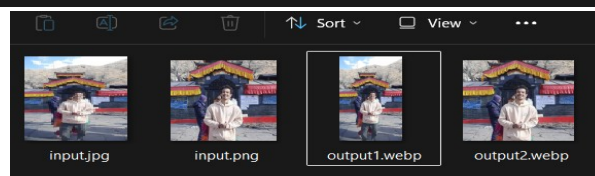
cwebp input.png -o output2.webp

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>cwebp input.jpg -o output1.webp
Saving file 'output1.webp'
File:      input.jpg
Dimension: 1125 x 2000
Output:    365532 bytes Y-U-V-All-PSNR 38.35 46.50 46.73   39.80 dB
          (1.30 bpp)
block count:  intra4:      6725  (75.77%)
               intra16:   2150  (24.23%)
               skipped:    495  (5.58%)
bytes used:  header:        455  (0.1%)
               mode-partition: 32436 (8.9%)
Residuals bytes | segment 1 | segment 2 | segment 3 | segment 4 | total
macroblocks:   |      13% |      15% |      40% |      31% | 8875
quantizer:     |      36 |      31 |      25 |      18 |
filter level:  |      11 |       6 |      11 |      21 |

C:\Users\Dell\Pictures\Saved Pictures\video>cwebp input.png -o output2.webp
Saving file 'output2.webp'
File:      input.png
Dimension: 340 x 323
Output:    24980 bytes Y-U-V-All-PSNR 36.70 41.15 41.09   37.74 dB
          (1.82 bpp)
block count:  intra4:       458  (99.13%)
               intra16:     4    (0.87%)
               skipped:     0    (0.00%)
bytes used:  header:        163  (0.7%)
               mode-partition: 2457  (9.8%)
Residuals bytes | segment 1 | segment 2 | segment 3 | segment 4 | total
macroblocks:   |      18% |      26% |      34% |      23% | 462
quantizer:     |      36 |      29 |      23 |      16 |
filter level:  |      11 |       6 |       4 |       2 |
```

Name	Date	Type	Size	Tags
 input.jpg	12/21/2023 8:46 PM	JPG File	293 KB	
 input.png	4/9/2026 9:40 PM	PNG File	250 KB	
 output1.webp	4/9/2026 9:42 PM	WEBP File	357 KB	
 output2.webp	4/9/2026 9:42 PM	WEBP File	25 KB	



b) Quality Levels (0–100)

cwebp -q 50 input.jpg -o output1.webp

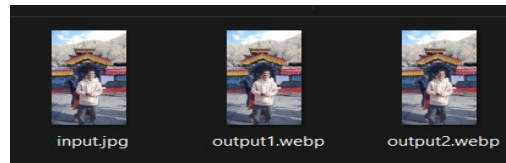
cwebp -q 90 input.jpg -o output2.webp

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>cwebp -q 50 input.jpg -o output1
.webp
Saving file 'output1.webp'
File:      input.jpg
Dimension: 1125 x 2000
Output:    288946 bytes Y-U-V-All-PSNR 35.60 44.03 44.33   37.07 dB
          (1.03 bpp)
block count:  intra4:      6562  (73.94%)
               intra16:   2313  (26.06%)
               skipped:    465  (5.24%)
bytes used:  header:        363  (0.1%)
               mode-partition: 30796 (10.7%)
Residuals bytes | segment 1 | segment 2 | segment 3 | segment 4 | total
macroblocks:   |      13% |      15% |      40% |      31% | 8875
quantizer:     |      52 |      46 |      37 |      28 |
filter level:  |      16 |      10 |      15 |      49 |

C:\Users\Dell\Pictures\Saved Pictures\video>cwebp -q 90 input.jpg -o output2
.webp
Saving file 'output2.webp'
File:      input.jpg
Dimension: 1125 x 2000
Output:    584322 bytes Y-U-V-All-PSNR 45.12 52.16 52.38   46.48 dB
          (2.08 bpp)
block count:  intra4:      7310  (82.37%)
               intra16:   1565  (17.63%)
               skipped:    480  (5.41%)
bytes used:  header:        771  (0.1%)
               mode-partition: 37957 (6.5%)
Residuals bytes | segment 1 | segment 2 | segment 3 | segment 4 | total
macroblocks:   |      13% |      15% |      40% |      31% | 8875
quantizer:     |      12 |      11 |       8 |       6 |
filter level:  |       4 |       2 |       2 |       0 |
```

Name	Date	Type	Size
input.jpg	12/21/2023 8:46 PM	JPG File	293 KB
output1.webp	4/9/2026 9:45 PM	WEBP File	283 KB
output2.webp	4/9/2026 9:45 PM	WEBP File	571 KB



c) Lossy vs Lossless

Lossy

cwebp -q 75 input.png -o output1.webp

Lossless

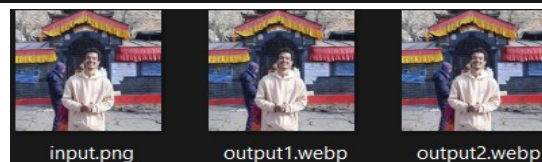
cwebp -lossless input.png -o output2.webp

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>cwebp -q 75 input.png -o output1
.webp
Saving file 'output1.webp'
File:      input.png
Dimension: 340 x 323
Output:    24980 bytes Y-U-V-All-PSNR 36.70 41.15 41.09   37.74 dB
           (1.82 bpp)
block count:  intra4:      458   (99.13%)
              intra16:     4   (0.87%)
              skipped:     0   (0.00%)
bytes used:  header:      163   (0.7%)
              mode-partition: 2457  (9.8%)
Residuals bytes|segment 1|segment 2|segment 3|segment 4| total
macroblocks:   |      18%|      26%|      34%|      23%|   462
quantizer:     |      36 |      29 |      23 |      16 |
filter level:  |      11 |       6 |       4 |       2 |

C:\Users\Dell\Pictures\Saved Pictures\video>cwebp -lossless input.png -o out
put2.webp
Saving file 'output2.webp'
File:      input.png
Dimension: 340 x 323
Output:    151506 bytes (11.04 bpp)
Lossless-ARGB compressed size: 151506 bytes
* Header size: 3917 bytes, image data size: 147563
* Lossless features used: PREDICTION CROSS-COLOR-TRANSFORM SUBTRACT-GREEN
* Precision Bits: histogram=3 prediction=3 cross-color=3 cache=0
```

Name	Date	Type	Size	Tags
input.png	4/9/2026 9:40 PM	PNG File	250 KB	
output1.webp	4/9/2026 9:48 PM	WEBP File	25 KB	
output2.webp	4/9/2026 9:49 PM	WEBP File	148 KB	



d) Resize Image

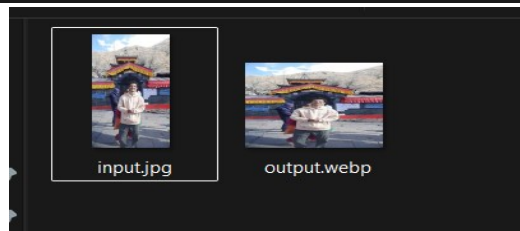
cwebp -resize 800 600 input.jpg -o output.webp

Output

```
C:\Users\Dell\Pictures\Saved Pictures\video>cwebp -resize 800 600 input.jpg
-o output.webp
Saving file 'output.webp'
File:      input.jpg
Dimension: 800 x 600
Output:    81634 bytes Y-U-V-All-PSNR 37.04 43.24 43.28  38.31 dB
(1.36 bpp)
block count:  intra4:      1697  (89.32%)
               intra16:    203  (10.68%)
               skipped:     62  (3.26%)
bytes used:  header:       219  (0.3%)
               mode-partition: 8691 (10.6%)
Residuals bytes segment 1 segment 2 segment 3 segment 4 total
macroblocks:   | 19% | 18% | 39% | 23% | 1900
quantizer:     | 36  | 30  | 23  | 17  |
filter level:  | 11  | 6   | 4   | 16  |

C:\Users\Dell\Pictures\Saved Pictures\video>
```

Name	Date	Type	Size
input.jpg	12/21/2023 8:46 PM	JPG File	293 KB
output.webp	4/9/2026 9:51 PM	WEBP File	80 KB



e) JPEG vs WebP Comparison

Feature	JPEG	WebP
File Size	Larger	Smaller
Quality	Good	Better at same size
Transparency	Not supported	Supported
Web Speed	Slower	Faster

Conclusion

FFmpeg is a powerful tool for multimedia processing that supports a wide range of operations including conversion, editing, filtering, and compression. The use of cwebp improves image optimization significantly, making both tools essential for modern multimedia and web development.

Lab 8: Write a program to implement a data compression algorithm.

Objective: To understand and implement basic data compression algorithms such as Run-Length Encoding (RLE) and Huffman Encoding.

Introduction: Data compression is a technique used to reduce the size of data for efficient storage and transmission. It can be lossless (no data loss) or lossy.

Algorithms Implemented:

Run-Length Encoding (RLE)

Concept: RLE compresses data by replacing repeated characters with a single character and its count.

Example:

Input: AAAABBBBCCDAA

Output: A4B3C2D1A2

Python Program:

```
def rle_encode(data):
    encoding = ""
    i = 0

    while i < len(data):
        count = 1
        while i + 1 < len(data) and data[i] == data[i + 1]:
            i += 1
            count += 1
        encoding += data[i] + str(count)
        i += 1

    return encoding
data = "AAAABBBBCCDAA"
print("Encoded:", rle_encode(data))
```

Huffman Encoding

Concept: Huffman Encoding is a lossless compression technique that assigns shorter codes to frequent characters and longer codes to less frequent ones.

Steps:

1. Calculate frequency of characters
2. Build a binary tree
3. Assign codes
4. Encode data

Python Program:

```
import heapq
from collections import Counter

class Node:
    def __init__(self, char, freq):
        self.char = char
        self.freq = freq
        self.left = None
        self.right = None

    def __lt__(self, other):
        return self.freq < other.freq

def build_huffman_tree(text):
    frequency = Counter(text)
    heap = [Node(char, freq) for char, freq in frequency.items()]
    heapq.heapify(heap)

    while len(heap) > 1:
        left = heapq.heappop(heap)
        right = heapq.heappop(heap)

        merged = Node(None, left.freq + right.freq)
        merged.left = left
        merged.right = right

        heapq.heappush(heap, merged)

    return heap[0]

def generate_codes(node, prefix="", codebook={}):
    if node:
        if node.char:

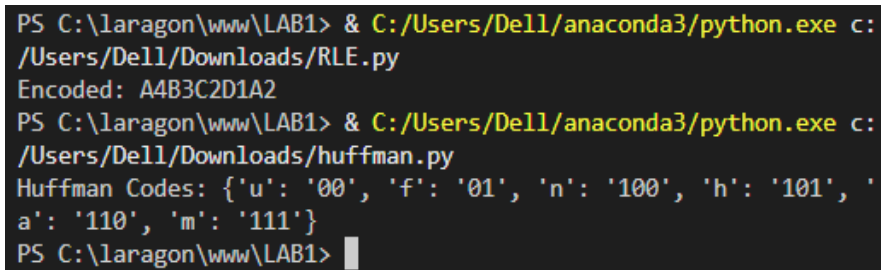
```



```
        codebook[node.char] = prefix
    generate_codes(node.left, prefix + "0", codebook)
    generate_codes(node.right, prefix + "1", codebook)
    return codebook
text = "huffman"
tree = build_huffman_tree(text)
codes = generate_codes(tree)

print("Huffman Codes:", codes)
```

Output:



```
PS C:\laragon\www\LAB1> & C:/Users/Dell/anaconda3/python.exe c:
/Users/Dell/Downloads/RLE.py
Encoded: A4B3C2D1A2
PS C:\laragon\www\LAB1> & C:/Users/Dell/anaconda3/python.exe c:
/Users/Dell/Downloads/huffman.py
Huffman Codes: {'u': '00', 'f': '01', 'n': '100', 'h': '101', '
a': '110', 'm': '111'}
PS C:\laragon\www\LAB1> █
```

Conclusion:

Run-Length Encoding is simple and effective for repetitive data, while Huffman Encoding provides efficient compression by using variable-length codes. Both techniques are important in data compression and real-world applications.

Project Report – Healthcare Booking App

Project Title

HealthConnect – Healthcare Booking Application UI/UX Design Using Figma

1. Introduction

HealthConnect is a healthcare booking mobile application designed to help users search for doctors, view doctor information, and book appointments digitally. The system provides a simple and organized platform where patients can access healthcare services quickly and conveniently.

This project focuses on creating a multimedia user interface using Figma. The design includes multiple interactive screens with proper navigation, visual consistency, multimedia elements, and usability principles. The main objective of the system is to improve the appointment booking process by reducing the difficulty of finding doctors and scheduling appointments.

The application combines UI/UX principles with multimedia elements such as icons, images, typography, color schemes, and interactive transitions to provide an effective user experience.

2. Objective

The main objectives of this project are:

- To create a modern and user-friendly healthcare application interface.
- To simplify doctor appointment booking.
- To provide easy navigation between different screens.
- To maintain visual consistency using proper color schemes and layouts.
- To improve accessibility and usability for users.
- To demonstrate multimedia integration using Figma design components.

3. Target Users

The healthcare booking application is intended for the following users:

- **Patients:** People who need to book doctor appointments quickly and efficiently.
- **Working Individuals:** Users with busy schedules who prefer digital appointment booking instead of visiting hospitals physically.

- **Elderly Users:** People who require a simple and organized interface to access healthcare services.
- **Urban Smartphone Users:** Users are comfortable with mobile applications and digital healthcare services.

4. Design Elements Used

The interface design uses several multimedia and UI elements to improve usability and visual appeal.

Color Scheme

A healthcare-based color palette was selected to create trust and comfort.

- **Primary Blue:** Represents trust, professionalism, and healthcare.
- **White Background:** Creates cleanliness and readability.
- **Light Gray Sections:** Used to separate content.
- **Green Accent:** Used for confirmation and success actions.

Typography

Simple and readable typography was used to improve readability.

- **Heading Font:** Bold and clean style
- **Body Font:** Simple and minimal
- **Consistent font sizes** for hierarchy

Multimedia Elements

The project includes:

- Doctor images
- Medical icons
- Search bar icons
- Interactive buttons
- Rounded cards and shadows

5. Description of Interfaces

The application includes five major screens.

a. Login / Signin Screen

This screen allows users to log into the healthcare application using their credentials.

Features:

- App logo
- Email or phone input field
- Password field
- Login button
- Signup option

Purpose: This screen provides secure access to the application.

b. Home Screen

The home screen acts as the dashboard of the application.

Features:

- Search bar for doctors
- Doctor categories
- Popular doctor cards
- Doctor images, names, and specialties

Purpose: Allows users to browse available doctors and search for specialists.

c. Doctor Detail Screen

This screen appears after selecting a doctor from the home screen.

Features:

- Doctor profile image
- Doctor name and specialty
- Experience and rating
- About doctor section
- Schedule Appointment button

Purpose: Provides detailed information about the selected doctor.

d. Appointment Booking Screen

This screen allows users to choose an appointment date and time.

Features:

- Calendar selection
- Time slot options
- Confirm appointment button

Purpose: Allows users to book appointments conveniently.

e. Confirmation Screen

This screen appears after a successful booking.

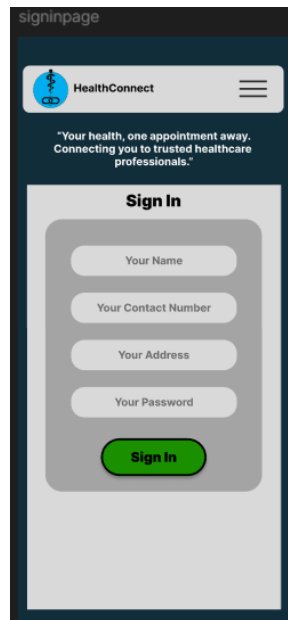
Features:

- Confirmation message
- Appointment details
- Return to home button

Purpose: Confirms successful booking and provides appointment information.

6. Screenshots of Designed Interfaces

1. Login Screen



Description:

This screen allows users to log into the HealthConnect application using email or password credentials. It includes input fields, a login button, and a signup option.

2. Home Screen



Description:

The home screen displays a search bar, doctor categories, and featured doctor cards. Users can browse doctors and select one to view details.

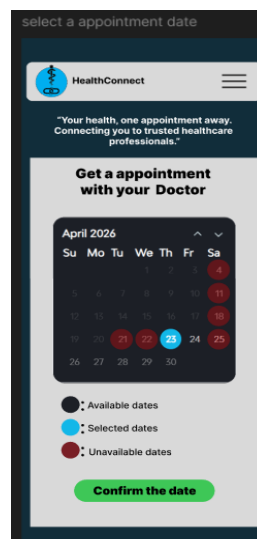
3. Doctor Detail Screen



Description:

This screen provides detailed information about the selected doctor including experience, specialty, and appointment scheduling option.

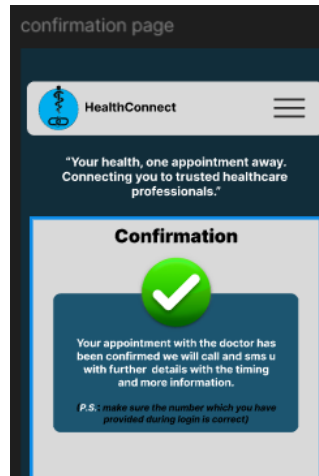
4. Appointment Booking Screen



Description:

Users can select available dates and time slots to schedule appointments.

5. Confirmation Screen



Description:

This screen confirms that the appointment has been booked successfully.

7. Prototype Interaction

The prototype was created using Figma's prototyping tools.

Navigation Flow

1. Login Screen → Home Screen
2. Home Screen → Doctor Detail Screen
3. Doctor Detail Screen → Appointment Booking Screen
4. Appointment Booking Screen → Confirmation Screen
5. Confirmation Screen → Home Screen

Prototype Features

- Button click interactions
- Screen transitions
- Connected navigation flow
- Interactive user journey simulation

The prototype creates a realistic application experience by allowing users to navigate through the booking process.

8. Conclusion

The HealthConnect healthcare booking application was designed to provide a simple and organized healthcare appointment system. The project successfully demonstrates multimedia UI design principles using Figma.

The application includes multiple screens, consistent layout design, visual hierarchy, multimedia integration, and interactive navigation. The design follows good usability practices while maintaining an attractive and professional appearance.

This project helped improve understanding of UI/UX principles, Figma prototyping, multimedia design components, and interactive application flow.

9. Figma Link

Design link: <https://www.figma.com/design/clYepdzq94MgYSrcYyqlkK/Untitled?node-id=0-1&t=6RGVv9oxiG4SR2df-1>

10. Prototype Link

Prototype link: <https://www.figma.com/proto/clYepdzq94MgYSrcYyqlkK/Untitled?node-id=5-4&p=f&t=6RGVv9oxiG4SR2df-0&scaling=scale-down&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=1%3A2>